Isometric Project Design Document

Version 1.7

This PDD template follows the Isometric Standard. Please refer to the Standard for more information on how PDDs are used as part of the procedures of the Isometric Registry.

Section A – Key Project Data

A.1 Title
Please provide the title of your Project. This will be displayed as your Project's name on the Isometric Registry and all related documentation.
NULIFE GreenTech Ontario Avenue Bio-Oil Sequestration

A.2 Description
Please provide a brief description (50-100 words) of your carbon removal Project. A more detailed written technical description must be provided later.
NULIFE GreenTech utilizes patented Hydrothermal Liquefaction (HTL) technology to convert local waste biomass into bio-oil. The bio-oil is transported to an operating third party injection site for sequestration into an underground salt cavern.

A.3 Project Location(s)
• *Please submit at least one Address and/or specific geo-coordinates for the project.* • *You may submit multiple Project locations – please specify what role each location plays in the Project.*
HTL Conversion Site: 1725 Ontario Avenue, Saskatoon, SK, Canada, S7K 1S9 Injection Site: LSD: [redacted]

A.4 Project Participants
Please provide a complete list of organizations participating in the project, with a contact person for each organization. (*Please duplicate the below rows for each additional organization you wish to add)*
Company registration number (*Unique business identification number in your country of*

registration): 101298013
Organization Name: NULIFE GreenTech Inc.
Organization Address: 1725 Ontario Ave, Saskatoon, SK, Canada, S7K 1S9
Contact Person: Kieran Lewko
Contact Email Address: KieranL@nulifegroup.com
Contact Phone Number: 306-500-5010 ext. 226
Organization role in Project: Project Owner, HTL technology provider, CDR credit sales.

Company registration number (*Unique business identification number in your country of registration*): [redacted]

Organization Name: [redacted]
Organization Address: [redacted]
Contact Person: [redacted]
Contact Email Address: [redacted]
Contact Phone Number: [redacted]
Organization role in Project: Injection partner.

A.5 Legal ownership of carbon removal claims
Please provide reasoning and evidence for legal ownership over the rights to all removals that will be claimed from the Project and refer to Section 3.1 “Ownership” of the Isometric Standard.
NULIFE GreenTech is the project owner and the only organization with ownership of the carbon removals associated with this project. Carbon ownership documentation associated with our biomass feedstock suppliers and the injection site are included in the attachments for this PDD.

A.6 Technical description of Project activity
Please provide a brief technical description of your carbon removal Project activity in accessible language. This should include information on facilities and equipment, the age and average lifespan of equipment, and all further information essential to understanding how carbon removal is achieved by the Project.

NULIFE's Hydrothermal Liquefaction (HTL) process is a continuous thermo-chemical process that breaks down and liquefies high-moisture organic materials. The exothermic reaction occurs under high pressure (2,900 – 3,700 psig) and temperature (290 - 375°C), using water as a catalyst to reform the biomass carbon into long-chain hydrocarbons (bio-oil).

NULIFE's facility is located at 1725 Ontario Avenue in Saskatoon, Saskatchewan, Canada. At this facility, multiple types of biomass feedstocks can be treated. NULIFE focuses on food processing and manufacturing wastes, such as effluent sludges, oat hulls, and other types of problematic wastes. NULIFE sources feedstocks for this project from Saskatoon and area. Once these feedstocks are converted by the HTL process, the resulting bio-oil is sent off to be injected into an underground salt cavern.

In 2022, the design process commenced for the Saskatoon Plant. In 2023, the first single commercial HTL processor system was built. In 2024, the Saskatoon Plant completed an expansion. A 2nd processor was added. This provided a 100% expansion to the production capacity.

In Q1 2025, the facility expanded to four commercial scale processors.

Increasing capacity to commercial scale, which is dubbed a "module", involves building a total of 12 processors, which are efficiently set on a steel rack to utilize vertical space. The first commercial module will be operational at our existing Ontario Avenue location in Q4 2025. The Saskatoon Plant is expected to operate until 2050.

Carbon Removal Steps:

1. Dry biomass waste is delivered and stored on site. Wet biomass is delivered and stored in 10,000L storage tanks.
2. The wet biomass feedstock is pumped and stored into holding tanks in preparation for the HTL process. The slurry can be mixed with additional dry biomass and/or water to obtain the desired solid content or can be processed "as received".
3. The slurry mixture moves through a series of pumps into the processor. The primary pump generates the necessary pressure for the HTL reaction (2,900 – 3,700 psig).
4. Once pressurized, the slurry travels through the HTL processor. The current processor heat source is electric heaters. The process operates at 290 °C to 375 °C.
5. After the reaction takes place, the effluent material discharges through a pressure letdown valve.
6. The effluent products are released into the flash tank. The bio-oil (with entrained biochar solids) and effluent water are transferred into the external bio-oil storage tank. The bio-oil accumulates until there is sufficient volume for a shipment to the injection site.
7. The wastewater collected from the separation of the bio-oil is transferred into the external water storage tank. In the Saskatoon plant, this wastewater can be recycled into the front of the process or sent for safe treatment/disposal offsite.
8. From the flash tank, the gas phase passes through an activated carbon canister and is then vented into the atmosphere. The resulting gas phase has been validated to be 99% biogenic CO_2, which will be released back into the atmosphere via the vent stack. Currently, this CO_2 is not captured. In the future it could be collected and repurposed for industrial applications to offset fossil CO_2 or injected underground for carbon dioxide removal.
9. The bio-oil liquid is transported to the injection site by a third-party road transport

company.
10. Upon arrival, the bio-oil is offloaded by the transport company at the injection site offloading pad. The truck offloads the bio-oil into the injectant tank through a connected hose.
11. The site operator then pumps the bio-oil underground into the salt cavern.
12. As bio-oil is injected, it displaces existing brine within the cavern. Brine is removed and is stored in brine tanks. If solid material is identified in the brine tank, it is removed and re-injected into the salt cavern.
13. Clean brine is discharged into a permitted brine disposal well onsite.

A.7 Declaration of exclusive registration	
Please confirm that your Project may only claim credits for activities that are exclusively registered with the Isometric Registry.	
x	I confirm the Project for which I aim to generate credits under the Isometric Standard is not registered with any other voluntary or compliance scheme.

A.8 Public funding
Please describe briefly whether your Project has received any public funding, e.g., grants or subsidies
Our Saskatoon Ontario Avenue project has received Canadian Federal & Provincial Grant funding.

A.9 Estimated carbon removal capacity	
Please give an estimate of the net carbon removal capacity of this project in the coming years (metric tonnes)	
Year	**Estimated carbon removal capacity (metric tonnes)**
2025	1,000
2026	4,500
2027	4,500
2028	4,500
2029	4,500
2030	4,500

Section B – Protocol and Monitoring Data

B.1 Selected Certified Protocol	
Please select the Isometric Certified Protocol you wish to use for this Project. *Please note that, as per the Isometric Standard, you must use the latest available version of a Certified Protocol, unless a grace period has been explicitly specified by Isometric, whereby a former version of a Protocol may continue to be used for a defined time period.*	
Protocol Name: Bio Oil Geological Storage	
Protocol Version Number: V1.1	
x	I confirm that I am using the most recent available Certified Protocol version, or that a grace period has been explicitly specified allowing the use of an earlier protocol version.

B.2 Project Eligibility

Please explain why this Project is eligible under the selected Protocol.

As Per Section 4.0 (Applicability) of the Bio-oil Geological Storage protocol, this project is eligible because:

- Biomass feedstocks are eligible in accordance with the Biomass Feedstock Accounting module. Please see the Biomass Feedstock Information appendix for full details on each feedstock and demonstration of eligibility.
- Project tech uses Hydrothermal Liquefaction (HTL), which is a similar process to pyrolysis to convert biomass to bio-oil. Section 1.0 (Introduction) of the protocol discloses liquefaction as a thermal chemical process to produce bio-oil.
- Bio-oil is injected into an underground salt cavern for long term storage.
- The project is net negative, as per the Net Negativity tool in Isometric's Certify Platform.
- The project does no net harm to the environment and society, as per the Environmental and Social Impacts sections.
- The project meets the additionality criteria – See the below sections demonstrating Additionality for further details.
- Project durability is over 1,000 years, as per the durability assessment.
- Section 4.0 (Applicability) of the Bio-oil Geological Storage protocol states that the project must use a geological storage site in the United States. The verbiage of the Bio-oil/Biomass Storage in Salt Caverns Module has been updated in V1.1 to become more inclusive to non-US injection sites.
- The geological storage site is properly permitted under Canadian Provincial regulations. Based on the location, the relevant regulating body is the [redacted]. The site permitting includes an acceptable injectant list that covers bio-oil or equivalent.
- The project does not perform Enhanced Hydrocarbon Recovery (EHR).

B.3 Acknowledgement of responsibility for providing notification of changes to operations

Please confirm that you acknowledge responsibility for notifying Isometric of changes to operations which deviate from this submitted PDD

x	I acknowledge responsibility for notifying Isometric of any changes to operations.

B.4 Project Boundary

Please provide a description of the project boundary and complete the below table, including all GHGs considered across all Sources, Sinks and Reservoirs (SSRs) in both the Project and Baseline scenario. An appropriate level of detail on the scope of each individual GHG SSR that was studied should be clearly set out. Robust justification and appropriate evidence must be provided for any GHG SSRs that have been excluded from the project boundary. Evidence may be documented here or appended to the GHG Statement or PDD separately.

Additionally, you may also provide a diagram of the project boundary.

Description of Project boundary:

1. Carbon filled waste residues and sludges are sourced from third party waste haulers or generators.
2. Biomass is diverted from unsustainable disposal routes to the HTL site.
3. HTL site converts biomass carbon into carbon rich bio-oil.
4. Bio-oil is transported to an injection site.
5. Bio-oil carbon is permanently stored in an underground salt cavern.

All activities undertaken to permanently sequester carbon, at the point of taking ownership of the organic waste, are included in the system boundary. Activities within the system boundary include biomass transportation, HTL conversion, bio-oil (and other output) transport, injection and monitoring activities. As a part of these activities, emissions associated with the procurement of all necessary equipment to run the facility are included, considering standard lifespans. Direct and embedded (upstream) emissions in fuels, materials, consumables and infrastructure are included. Also included is the counterfactual use case of the organic waste biomass and any relevant replacement emissions.

Excluded from the system boundary are all upstream inputs to biomass cultivation and production (all inputs into growing the biomass itself, as well as transportation and other processing activities involved in creating the waste). These upstream inputs were excluded because the biomass taken is waste – not purpose-grown biomass – so these upstream activities occur without NULIFE's involvement.

Excluded from the system boundary are HTL tailgas biogenic Carbon Dioxide (CO_2) emissions released through the chimney. These are excluded because their emissions factor is 0 because it is biogenic CO_2.

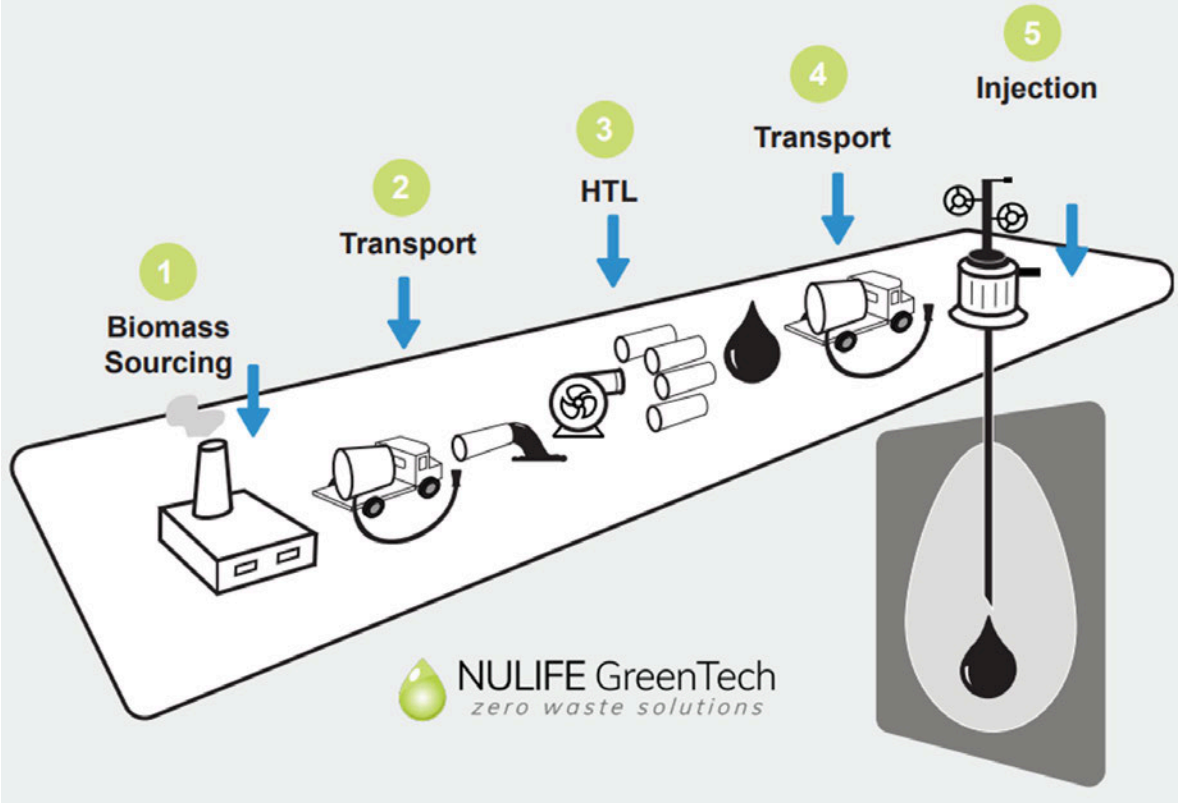
Excluded from the system boundary are the embodied emissions associated with the construction of the HTL conversion site building and the injection site. NULIFE' Saskatoon plant is located in a building that was built in the 1970s. The Injection site was first opened in the 1960s for natural gas storage. It was turned into a waste disposal site in the 1990s.

Neither site was developed for the purpose of this Carbon Dioxide Removal project.

Methane emissions that would have occurred in the counterfactual scenario of the waste decomposing without NULIFE's intervention are excluded.

Geographic Boundary: Biomass waste is sourced from within the geographic boundary that supports net-negative carbon removal. Typically, this waste is within a 40km radius of the Saskatoon Plant. This HTL facility is located in Saskatoon. The injection site facility is located in [redacted] Location is considered when selecting grid emission factors for energy consumption.

[Optional] Project boundary diagram:



Baseline / Project	**Carbon Flux / Emission Source**	**Included/ excluded from project boundary?**	**Greenhouse Gas(es)**	**Justification/description**
Baseline	Counterfactual emissions. Fuel use from feedstock sourcing	Excluded	CO_2, CH_4, NO_2	No additional fuel usage on part of the waste supplier due to NULIFE activities. The feedstocks are waste streams/by-products of commercial activities and if these activities took place, they would have taken place regardless of the project. These emissions are 0.

Baseline	Counterfactual emissions. Electricity use for feedstock sourcing	Excluded	CO_2, CH_4, NO_2	No additional electricity usage on part of the waste supplier due to NULIFE activities. The feedstocks are waste streams/by-products of commercial activities and if these activities took place, they would have taken place regardless of the project. These emissions are 0.
Baseline	Counterfactual emissions. Replacement of feedstock	Included	CO_2, CH_4, NO_2	Multiple feedstocks in the PDD will result in varying Replacement emissions. When relevant, NULIFE will demonstrate these emissions. For example, the dairy Dissolved Air Flotation (DAF) sludge feedstock usage considers fertilizer replacement emissions for landspreading activities. When relevant, NULIFE will demonstrate that these emissions can be considered 0 due to eligibility criteria in the Biomass Accounting Module.
Baseline	Counterfactual emissions. Temporary carbon storage from feedstock	Included	CO_2, CH_4.NO_2	NULIFE will demonstrate the quantity of stored carbon, on a per feedstock basis. When relevant, NULIFE will demonstrate that these emissions can be considered 0 due to eligibility criteria in the Biomass Accounting Module.
Project	Fuel Use from feedstock transport	Included	CO_2, CH_4, NO_2	Fuel used from transport activities is calculated in LCA and utilizes an appropriate emissions factor to consider CO_2, CH_4, NO_2. The emissions factors used for truck transport include embodied emissions from truck manufacture and fuel production and the impacts of fuel use, on a tonnekm basis.
Project	Fuel Use from bio-oil transport	Included	CO_2, CH_4, NO_2	Fuel used from transport activities is calculated in LCA and utilizes an appropriate emissions factor to consider CO_2, CH_4, NO_2. The emissions factors used for truck transport include embodied emissions from truck manufacture and fuel production and the impacts of fuel use, on a tonnekm basis.
Project	Fuel Use from wastewater transport	Included	CO_2, CH_4, NO_2	Fuel used from transport activities is calculated in LCA and utilizes an

				appropriate emissions factor to consider CO_2, CH_4, NO_2. The emissions factors used for truck transport include embodied emissions from truck manufacture and fuel production and the impacts of fuel use, on a tonnekm basis.
Project	Fuel Use from consumable (Sodium Hydroxide and Hydrogen Peroxide from the same supplier) transport.	Included	CO_2, CH_4, NO_2	Fuel used from transport activities is calculated in LCA and utilizes an appropriate emissions factor to consider CO_2, CH_4, NO_2. The emissions factors used for truck transport include embodied emissions from truck manufacture and fuel production and the impacts of fuel use, on a tonnekm basis.
Project	Fuel Use from consumable (activated carbon) transport.	Included	CO_2, CH_4, NO_2	Fuel used from transport activities is calculated in LCA and utilizes an appropriate emissions factor to consider CO_2, CH_4, NO_2. The emissions factors used for truck transport include embodied emissions from truck manufacture and fuel production and the impacts of fuel use, on a tonnekm basis.
Project	Embodied Emissions for Transport	Included	CO_2, CH_4, NO_2	Mass x distance emissions factor selected takes into consideration embodied emissions of the vehicle type and expected useful life. The emissions factors used for truck transport include embodied emissions from truck manufacture and fuel production and the impacts of fuel use, on a tonnekm basis.
Project	Electricity Use for HTL Conversion - Primary Power Meter	Included	CO_2, CH_4, NO_2	CO_2 included as the primary emission from electricity consumption. CH_4 and NO_2 included for completeness as may also be released during electricity generation.
Project	Electricity Use for HTL Conversion - Non power meter equipment log	Included	CO_2, CH_4, NO_2	CO_2 included as the primary emission from electricity consumption. CH_4 and NO_2 included for completeness as may also be released during electricity generation.
Project	Electricity Use for HTL Conversion - sampling equipment (Carbon wt% & GC gas analysis)	Included	CO_2, CH_4, NO_2	CO_2 included as the primary emission from electricity consumption. CH_4 and NO_2 included for completeness as may also be released during electricity

				generation.
Project	Electricity Use for Injection Site	Included	CO_2, CH_4, NO_2	CO_2 included as the primary emission from electricity consumption. CH_4 and NO_2 included for completeness as may also be released during electricity generation.
Project	Embodied Emissions for HTL Conversion Site Equipment	Included	CO_2, CH_4, NO_2	The Mass based method is used for calculating the emissions associated with equipment for the HTL conversion site. Materials include steel, stainless steel and polyethylene plastic. Appropriate cradle to grave emissions factors are used to determine the CO2e based on the mass of each material.
Project	Embodied Emissions for Injection Site Equipment	Included	CO_2, CH_4, NO_2	The Mass based method is used for calculating the emissions associated with equipment for the HTL conversion site. Materials include steel, stainless steel and polyethylene plastic. Appropriate cradle to grave emissions factors are used to determine the CO2e based on the mass of each material.
Project	Embodied Emissions for HTL Conversion Site Construction	Excluded	CO_2e	Excluded because the site was not built for this project.The site was built in 1970 for industrial use.
Project	Embodied Emissions for Injection Site Construction	Excluded	CO_2e	Excluded because the site was not built for this project. The site was built in 1960 for natural gas storage.
Project	Embodied Emissions for HTL Conversion Site consumables.	Included	CO_2, CH_4, NO_2	Emissions factor for consumables are taken from cradle-to-gate LCA datasets, which include all impacts from raw materials extraction, transport, production, and waste disposal.
Project	Carbon Content of Injected bio-oil	Included	CO_2e	Primary negative emission from process. At least three carbon wt % samples are taken per HTL production batch. Each sample is tested in duplicates. Mass of bio-oil is determined via mass measured per HTL production batch.
Project	CH_4 content of HTL tailgas	Included	CO2e	wt% CH_4 of the tailgas is tested via GC gas analysis. The mass is determined through material balance. CH_4 tailgas released into the atmosphere via the chimney use an emissions factor to convert CH_4 to CO_2e using the appropriate emissions factor.

Project	CO_2 content of HTL tailgas	Excluded	CO_2	wt% CO_2 of the tailgas is tested via GC gas analysis. The mass is determined through material balance. CO_2 tailgas released into the atmosphere via the chimney uses an emissions factor of 0 because it is biogenic CO_2.
Project	Staff Travel - Transport Emissions from Transport of test samples to lab	Included	CO_2, CH_4, NO_2	Transport events are recorded. The transport distance (from site to lab to site) can be determined using Google Maps. Emissions are calculated using the Distance based method using an appropriate emissions factor for gasoline car transport include embodied emissions from vehicle manufacture and fuel production and the impacts of fuel use, on a km basis.
Project	Staff Travel and Surveys	Included	CO_2, CH_4, NO_2	Staff travel related to operators round trips to work are included. Distance based method using an appropriate emissions factor for gasoline car transport include embodied emissions from vehicle manufacture and fuel production and the impacts of fuel use, on a km basis. No surveys have been completed for this project and is considered out of scope.
Project	Sample Delivery - Injection Site to Nulife	Excluded	CO_2, CH_4, NO_2	The transport emissions associated with the delivery of brine and gas samples from the Injection Site to NULIFE for analytical testing is not included. The justification is as follows: Purolator (Canadian road courier) is used for transport, so the transport movement was already taking place regardless.The packages are picked up using the online portal booking system. Due to the extremely small mass (< 0.25kg) of the samples, the marginal emissions for moving a package is nearly zero. No significant new emissions occur since the package does not require an additional trip, vehicle, or route diversion.

Project	Equipment Transport from supplier to end use Site (i.e. Nulife site)	Included	CO_2, CH_4, NO_2	The emissions factors utilized for equipment are cradle to gate emissions factors. Emissions from transport between the material supplier and the end use site are not included. Transport emissions between the supplier location and NULIFE's site are calculated based on a round trip and the mass of the steel, stainless steel and plastic.
Project	Wastewater Disposal	Included	CO_2, CH_4, NO_2	When wastewater is disposed of in , we account for the emissions using an appropriate emissions factor.

Note: Carbon Flux/emissions sources will be subject to the Isometric GHG Accounting (V1) Materiality Assessment (Section 4). Carbon Flux/emission sources that are labelled "included" in the table above may be omitted from removal data, as per the requirements of section.

B.5 Baseline scenario

Please describe the baseline scenario of what would have happened if your Project did not take place (refer to Section 2.5.2 "Baselines" of the Isometric Standard and the requirements outlined in the relevant Protocol). Projects will only be credited for Removals above this counterfactual baseline.

Biomass Feedstocks

NULIFE sources organic wastes that, in their baseline counterfactual, would have primarily decomposed above ground, re-releasing the CO_{2e} back into the atmosphere. No credit is given for any avoided methane emissions from baseline scenarios such as the landfilling of waste biomass.

Each feedstock used in the project undergoes requirements in the Biomass Feedstock Accounting module to verify eligibility. Counterfactual scenarios related to replacement emissions (Section 3.2) are considered and vary per feedstock. Carbon Storage from Feedstock (Section 2.2) scenarios are considered and vary per feedstock.

More details on each feedstock can be found in the Biomass Feedstock Accounting PDD Appendix.

Injection Site

The Injection Partner is an active waste management company, with various Canadian underground injection assets. The injection site is fully operational, with a footprint that does change from NULIFE's project activities. These caverns were used in the 1960s for natural gas storage. In the 1990s they were sold and repurposed into waste disposal caverns.

In order to ensure a conservative estimate of removals, the embodied emissions of all project equipment and work on the site, as well as emissions associated with energy consumption, is accounted for and amortized against net removals associated with NULIFE bio-oil. The Injection partner uses its salt cavern and associated infrastructure for underground waste emplacement irrespective of its relationship with NULIFE and participation in this project. NULIFE accounts for all project-specific emissions as a result

of emplacement at the injection site, including a fraction of their equipment embodied emissions in proportion with the use of the equipment for project emplacements.

B.6 Leakage Assessment

Please give a robust assessment of how you have considered potential increases in GHG emissions outside the defined Project boundary that occur as a result of the Project activity. Where the potential for such Leakage is identified, it must be quantified and deducted from the CO_2 Removals in accordance with the relevant Protocol. Please refer to Section 2.5.4 "Leakage" of the Isometric Standard.

NULIFE accounts for potential leakage across all stages of the project. Potential areas of leakage outside of the project boundary include:

Biomass - Market Leakage Associated with Feedstock Displacement

We target waste streams and do not disrupt the current model of waste transportation.

Due to our focus on waste biomass streams, we have greatly minimized the risk of leakage for this project. Our feedstocks are waste streams/by-products from existing industrial operations. The operations have primary sources of revenue from their primary products and the potential for shifts in market behavior from NULIFE's project is non-existent.

Our primary feedstocks are wastes—that would otherwise be managed through methods such as landfilling or land application. By diverting these waste streams, our project avoids emissions rather than shifting them elsewhere.

Market effects associated with biomass must be assessed on a case by case basis using the Isometric Biomass Feedstock Accounting Module. Feedstock details are discussed in the Biomass Appendix of this PDD.

In relation to the current feedstocks for this PDD:

- Bran waste's previous disposal method: landfill.
- Sludge waste's previous disposal method: stockpile with no alternative use.
- Oat Hulls previous disposal method: landspreading.

Biomass - Replacement Emissions

Replacement emissions are determined in compliance with Section 3.2 of the Biomass Feedstock Accounting Module.

Any quantification associated with leakage is done as part of the Biomass Feedstock Accounting Module. When emissions factors are required to calculate replacement emissions, they consider the total lifecycle of the replacement product, including transport. There is no risk of leakage beyond what would be quantified for replacement emissions in section 3.2.

In relation to the current feedstocks for this PDD:

- Bran waste: No replacement emissions.
- Sludge waste's previous disposal method: No replacement emissions.
- Oat Hulls: No replacement emissions due to volume acquired beyond the sustainable use rate, thus they are classified as true waste, as per Section 3.2 of the Biomass Feedstock Accounting Module.

Feedstock details are discussed in the Biomass Appendix of this PDD.

Geological Storage Site

There is no identified leakage risk associated with the injection site. The injection site involves an operational company whose activities do not change with the addition of bo-oil injection. There is no risk for activity shifting or market transformation from this project.

Salt caverns are deeply isolated from biological ecosystems. The functional use of the salt

caverns does not change from this project, thus the risk of shifts in ecosystem processes is not present.

B.7 Demonstration of Additionality

B.7.1 Demonstration of Additionality: Financial Additionality

Please describe and provide evidence for how your Project is financially additional. Refer to Section 2.5.3. "Additionality" of the Isometric Standard and the requirements outlined in the relevant Protocol.

The outcome of the project, which is the net removal of carbon, would not have occurred in the absence of a carbon removal market. This is supported by the high cost of the process, including the HTL operations, transport, and injection in a salt cavern.

The revenue from the sale of Carbon Dioxide Removal (CDR) credits becomes instrumental in making the project financially viable. Alone, a financial model which may include feedstock tipping fees and government grants as the only revenue would not be enough to cover the costs of the process. Currently, NULIFE does not receive a paid tipping fee for any of the feedstocks included in this PDD. The sequestration of carbon is therefore dependent on the issuance and sale of CDRs. Without carbon revenue, the project would not be viable, nor financially attractive to generate investment. This is validated by project financials. As per the Isometric Standard, an Internal Rate of Return (IRR) was calculated for the project by assessing the financials over 10 years, including a sensitivity analysis of 20% on all key inputs. The result of this analysis is a negative project IRR in all scenarios.

Project financials, with sensitivity analysis, are attached to validate financial additionality as per the Isometric Standard.

B.7.2 Demonstration of Additionality: Environmental Additionality

Please describe and provide evidence for how your Project is environmentally additional. Refer to Section 2.5.3. "Additionality" of the Isometric Standard and the requirements outlined in the relevant Protocol.

NULIFE's Saskatoon project is net negative after subtracting the counterfactual CO_2 removal and all project GHG emissions from project CO_2 removals. The project therefore qualifies as environmentally additional as defined by the Isometric Standard. This can be validated by assessing the Net Negativity work completed in Isometric's Certify Platform.

B.7.3 Demonstration of Additionality: Regulatory Additionality

Please describe and provide evidence for how your Project is regulatorily additional. Refer to Section 2.5.3. "Additionality" of the Isometric Standard and the requirements outlined in the relevant Protocol.

NULIFE's Saskatoon project is not required by any regulatory, policy or other legal requirement. The purpose of the project is to sequester carbon that would have otherwise been returned to the atmosphere in order to generate credits that can be sold in the voluntary market. The project qualifies as additional based on the Isometric Standard.

B.7.4 Demonstration of Additionality: Common Practice Additionality

Projects must demonstrate that activities similar to the activities of the proposed project are not common practice as per Section 2.5.3.1 Common Practice Analysis. Please select one of the two options below and provide an explanation if needed.

X	*This Project is in a pathway which is **below** technology readiness level (TRL) 8 or 9, according to the definition in the Standard section above, and therefore may be considered as demonstrating common practice additionality without further analysis, due to the level of maturity of the relevant technology.*
•	*This Project is in a pathway which has a technology readiness level **above** 7, according to the definition in the Standard section above, and therefore is submitting a full common practice analysis, describing the applicable geographic area and assessment of similar activities, technologies, services or practices as described in the Isometric Standard.*

The first option is selected. As per Section 2.5.3.1, our project is in the section BiCRS/Bio-CCS with an acceptable TRL level, thus the condition is satisfied. We are demonstrating common practice additionality without further analysis due to the level of maturity of the relevant pathway.

B.8 Durability Assessment

Please confirm that the durability of your chosen sequestration pathway is at least 1000 years, in accordance with the Isometric Standard, and provide justification for how the Project adheres to the durability requirements outlined in the selected Protocol. This may include references to published literature or internal research. You may further expand on the Monitoring approach used to support the claimed Durability assessment in the "Overview of monitoring for durability" section of the next section.

x	I confirm that the durability of carbon removed through this Project is at least 1000 years, in accordance with the Isometric Standard.

Salt caverns are created in existing salt domes, many of which were formed ~30 million years ago. A salt cavern is a storage cavern that has been constructed by dissolving an underground formation which is made up primarily of salt. The cavern is formed through the process of solution mining. Fresh water is pumped down a borehole into the salt. Some of the salt is dissolved leaving a void and the water, now saline, is pumped back to the surface. The process continues until the cavern has been washed to the desired size. Once created, a salt cavern offers an underground waste storage vessel with very high deliverability.[1] Salt caverns, like the ones in Alberta and Saskatchewan, are at depths of 1,200 to 1,500 meters underground[2]. Their locations are determined while considering proximity to human centers, future land-use in the area while considering risks associated with leakage and ground water contamination[3]. The degree of environmental security associated with salt cavern storage is tremendous[4]. Once a salt cavern is built, they are filled with brine and materials are injected as a slurry with water and brine. Salt is an impermeable formation, thus making it an effective method for securely storing waste materials without risk of leakage into surrounding environments. When filled, the cavern will undergo slow closure and the solids will be permanently entombed (buried) in the low permeability salt[5].

Dissolved salt caverns represent an approach to geological sequestration of carbon. In this approach, a large volume cavern is created by solution mining and then filled with carbon containing substances before it is sealed indefinitely (>10,000 years)[6].

NULIFE stores carbon rich bio-oil in sealed salt caverns for permanent geological storage. Neither leakage nor re-emissions of the bio-oil and its carbon content is expected. Durability is expected to exceed much further than 10,000 years. This expectation is based on a combination of direct measurement, and via scientifically falsifiable hypotheses that there is no alternative destination for CO_2 storage other than the salt cavern in question.

This method leverages the stability and geologic conditions of salt caverns, which have been used historically for storing natural gas and other substances due to their impermeability and stability. Salt caverns have been identified in literature as a viable location for permanent storage of CO_2[7]. A review in the Energies journal highlights the benefits of HTL bio-oil polymerizing under pressure, reducing its reactivity and mobility, which is ideal for long-term storage in geological formations like salt caverns. This solidification process essentially locks the carbon in place, preventing it from re-entering the atmosphere.[8]

Through appropriate selection of waste biomass sludges and/or residue feedstocks that would otherwise decompose or combust, the conversion of biomass to bio-oil and the permanent geological storage of the embodied carbon is a net removal from the atmosphere.

<u>Site specific information on Durability</u>

Located near [redacted], the site contains 5 underground salt caverns. These salt caverns were originally developed in the 1960s for natural gas storage, a proven, high-integrity storage application.

Source: [redacted]

Source: [redacted]

[redacted]

These caverns were formed within deep bedded salt formations of the Prairie Evaporite Formation, a thick, regionally extensive salt layer across the Prairie Basin.

Source:Prairie Evaporite.

The caverns have maintained full integrity with no recorded events such as pressure loss, casing faults, or structural issues for over 65 years.

Presently, the site operates as a waste disposal facility with two operational caverns and three caverns that are yet to be re-commissioned for disposal/sequestration.

As per the [redacted] permitting requirements, the salt caverns must comply with code: CSA (Canadian Standards Association) Z341.2 Salt Cavern Storage. The code includes testing associated with:

- Material requirements,
- Well completion and conversion,
- Location of underground storage facility,
- Design and development,
- Development and construction,
- Surface facilities (w/ emergency shutdown valves),
- Operations and maintenance,
- Monitoring and measurement,
- Safety,
- Closure: Plugging, abandonment and site restoration.
- CSA Z341 SERIES:22 | Product | CSA Group

Compliance with CSA Z341.2 ensures site-specific durability by establishing rigorous standards for the design, construction, monitoring, and closure of salt caverns used for fluid storage. The standard requires comprehensive evaluation of the local geology, cavern integrity, and wellbore sealing to prevent any potential migration of stored materials. It mandates ongoing monitoring (pressure, mechanical integrity,etc.) and robust closure procedures (cementing and permanent abandonment of wells) to isolate the cavern from surface pathways.

The injection site monitors the displaced brine returned from the cavern for any waste, which is then filtered out and recycled back into the injection stream. As per permitting requirements, salt cavern operations require close looped systems for brine streams. If material is removed in the brine, the material must be collected and re-injected into the disposal cavern. If there is a spill onsite, the operator must dig out the spill and place the entire spill material into the disposal cavern. These permitting requirements help reduce reversal risk.

As per the monitoring plan section of the PDD, discharge brine and gas testing will be implemented to assess carbon reversal risk beyond what is required under permitting requirements.

Four site specific attachments are provided as supplementary material to this Durability section. The attachments cover: site specific facility descriptions, third party engineering review, integrity of the caverns and more.

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[1] Salt Cavern Storage Well Basics · CG Engineering · Calgary, Alberta

[2] Salt Solution Caverns for Petroleum Industry Toxic Granular Solid Waste Disposal | SPE/ISRM Rock Mechanics in Petroleum Engineering | OnePetro

[3] Outside the Box – Salt Caverns - Apollo Mapping

[4] Deep geological placement of wastes | Wat On Earth | University of Waterloo (uwaterloo.ca)

[5] Deep geological placement of wastes | Wat On Earth | University of Waterloo (uwaterloo.ca)

[6] (PDF) Sequestration of CO in Salt Caverns (researchgate.net)

[7] (PDF) Sequestration of CO in Salt Caverns (researchgate.net)

[8] https://www.mdpi.com/2073-4344/12/12/1621

B.9 Monitoring Plan

Please specify whether this Project will follow the default Monitoring Plan, following the requirements specified in the "Monitoring Plan Requirements" appendix of the relevant Protocol and its associated modules (Option A), or whether a Monitoring Plan will be submitted which may contain additional or modified monitored parameters (Option B).

Additionally, please give an overview of how monitored parameters and/or models will be used to support the assessment of durability. You may also provide any additional information here about how Project monitoring will be conducted.

●	A – I confirm this Project will follow the default Monitoring Plan, following the requirements specified in the "Monitoring Plan Requirements" appendix of the relevant Protocol and its associated modules
x	B – I have submitted a Monitoring Plan, which meets the requirements specified in the "Monitoring Plan Requirements" appendix of the relevant Protocol and its associated modules, and which describes any additional or modified parameters monitored which deviate from the default Monitoring Plan

Overview of how monitored parameters or models will be used to support assessment of durability. If Option B was selected, please give details of how the monitoring plan adds to or differs from the default monitoring requirements outlined in the Protocol:

[Optional] Additional information on Project monitoring:

Option B selected. The following data highlights how the monitoring plan slightly deviates from the default monitoring plan outlined in the protocol.

Monitoring Plan: Bio-Oil Geologic Storage module Monitoring Plan (Section 11.2)

Wt % of C in the bio-oil injectant

- As per section 7.4.1.1.1, 'Method A: measure every batch' has been selected. Bio-oil production batches are blended in NULIFE's external storage tank prior to shipment for injection.
- Carbon wt% is determined by elemental analysis (ASTM D5921 or equivalent) using the University of Saskatchewan Chemical Engineering Lab (deemed an in-house lab that is not ISO17025 accredited lab*).
- A minimum of 3 samples are taken per HTL production batch, as per the minimum samples per batch requirements in Section 7.4.1.1.1.
- Samples are taken from a well mixed allotment, including any solids.
- Samples are dried in an oven, to remove moisture, before preparation for elemental analysis.
- Samples are driven, bi-weekly, to the lab which is 6km away.
- The lab technician prepares the samples for elemental analysis, which runs bi-weekly.
- Each sample is tested for carbon wt% in duplicates and the average of the two samples is utilized for each of the three samples. Example:
    - S1: A: 75% Carbon, B: 76%. Average = 75.5%
    - S2: A: 75% Carbon, B: 75%. Average = 75%
    - S3: A: 74% Carbon, B: 75%. Average = 74.5%
    - Average of HTL production batch: 75%.
- In the case of carbon outliers (variance between the duplicate samples), the lab technician is asked to re-run samples to re-assess. The variance is +/- 5 wt%,. For example, if the duplicates resulted in A: 75 wt% and B: 70 wt%, the samples would be re-run.
- NULIFE's internal tracking program has a mechanism built in to flag carbon wt% outliers so they are rapidly identified. This is a secondary check beyond the first human check identified above.
- Often, NULIFE will test more than 3 samples per production run. Regardless of how many samples are tested, we take a weighted average of all samples assessed, which includes all duplicates. For example, if 4 samples were tested, there would be a total of 8 carbon data points for the relevant production run.
- The emissions factor of 3.67 CO_2e eq mass of C is used to convert bio-oil carbon into CO_2.

*An In-house lab is deemed acceptable by Isometric by supplying more information. Attachment "Carbon content lab test requirements v1.0" is provided for more details on the procedures of the equipment.

Total mass of bio-oil injectant

- Bio-oil Mass is determined by density measurement including a deduction for moisture content of the bio-oil.
- Mass of bio-oil is measured in kg and allocated per HTL production run.

- The density of bio-oil is determined via ASTM D4052 or equivalent through testing at the University of Saskatchewan Chemical Engineering Lab. A bio-oil density value is determined for bio-oil from each feedstock source.
- The moisture content (wt %) of bio-oil is determined by ASTM E-1756 through onsite testing. This is tested on three bio-oil samples from every HTL production run and the average of the three samples is utilized. The procedure includes an extra step past the initial drying process where the sample is dried for an additional hour after the first weighing. This extra drying takes place until the mass of the sample has not changed by more than 1% between drying. If the mass of the sample changes by more than 1% the process is repeated for an extra hour. This extra step is to ensure that all of the moisture has been taken out of the sample.
- The combination of operational data from multiple HTL production batches will result in a total mass of bio-oil for injection. For example: Trial A had 500kg and Trial B had 500kg, thus the total mass for CO_2e Stored calculations would be 1,000kg. The existing protocol implies the option to use the scale at the injection site for this calculation. Due to varying moisture levels in the bio-oil and the commercial impracticality of entirely removing the moisture, the injection site scale will record a higher mass and result in an over statement of CO_2e stored. For transport/injection related emissions, discussed elsewhere in the PDD, the moisture must be accounted for.

Tailgas - Mass flow rate of tailgas

- Tailgas mass flow is measured in kg per HTL production batch.
- Flow per HTL production run measured via material balance remainder analysis.

Concentration of CO_2 and CH_4 in tailgas

- The sample point for the gas sample is before the tailgas pass through the activated carbon canister. The activated carbon removes VOCs, while CO_2 and CH_4 pass through.
- Tailgas composition (CO_2 and CH_4) is measured using GC Analysis at the University of Saskatchewan Chemical Engineering Lab (deemed an in-house lab that is not ISO17025 accredited lab*).
- Total mass of the sample, and each unique gas, (grams) is determined by taking the mol % reading of each gas and multiplying it by the g/mol of the relevant gas.
- Wt% of CO_2 and CH_4 in the tailgas stream is determined by taking the mass of the gas and dividing it by the total mass of the total sample.
- GC analysis is performed, at minimum, per feedstock per unique operational parameter.
- An emissions factor of 0 is used for CO_2 emissions since it is biogenic CO_2.
- A relevant emissions factor is used for methane emissions on a mass of CO_2e per mass of methane basis.

*Attachment “Carbon content lab test requirements v1.0” is provided for more details on the procedures of the equipment.

Biomass conversion batch operation time

- Measured per HTL production batch in start up, operation and cooldown sections.
- Measured in hours.

Mass of fuel used in biomass conversion

- Currently the HTL system does not utilize fuel to convert biomass into bio-oil.
- When relevant, the fuel usage would be recorded per HTL production run.
- As per section 7.1 of the protocol, the building has out of scope usage such as research & development and corporate administration activities, thus the usage of monthly utility statements for natural gas usage is not used.

Electricity usage for biomass conversion

- Total electricity consumption (kWh) is measured per HTL production run.
- The electricity consumption is measured using a power meter that is connected to the primarily electrical equipment of the Saskatoon plant. If misc. equipment is used, that is not connected to the power meter, the usage is manually recorded and converted to kWh. The power meter accounts for +99% of kWh at the HTL site.
- Equipment that is connected to the power meter: processor heaters, pumping, valving, PLC, pressure/temperature/flow instrumentation, level sensors, Uninterruptible Power System (UPS).
- Equipment that uses plug in power and that is not connected to the power meter: oven for drying samples, forklift charger and an air compressor. For this equipment, manual calculations are recorded for kWh based on equipment specifications and recorded time used per HTL production run.
- Electricity consumption is calculated on a kWh per sample basis for Elemental Analysis (carbon wt%) and GC gas analysis.
    - Elemental analysis uses the equipment KW values and multiplies it by the hours to process a single sample to arrive at the kWh per sample value.
    - GC analysis uses the equipment Volts and amps values to determine the KW value. The KW value is multiplied by the hours to process a single sample to arrive at the kWh per sample value.
    - Based on the number of samples processed, and the kWh per sample value, we can calculate the emissions associated with electricity usage for sample equipment.
- As per section 7.1 of the protocol, the building has out of scope usage such as research & development and corporate administration activities, thus the usage of monthly utility statements for kWh is not used.

Electricity Emissions Factor

- Saskatchewan grid electricity emissions factor is used on a mass of CO_2e per kWh of electricity used basis.

Source: En84-294-2024-eng.pdf

Electricity Renewable Electricity Certificates (RECs)

- As per Section 5.1 of V1.2 Energy Use Accounting Module, our plant is deemed a non-intensive facility. The limit is 200 gWh per year or 200 million kWh. For reference, one hour of operation per HTL processor will utilize ~25kWh of electricity. Assuming 24/7 operation, 365 days per year with 10 processors, the total ends up at 2,190,000 kWh. This is 1% of the maximum allowable kWh.
    - Math: 35kwh x 10 processors x 24hr/day x 365 days/year = 2.19M kWh.
- RECs are acquired from SaskPower and used to offset the energy intensity of the grid electricity in Saskatchewan. This is through SaskPower's newly launched Renewable Subscription Service program. The renewable energy will be drawn from the same grid region from new facilities (<36 months), and the electricity will not be generated more than one year prior to the point of consumption by the project. Retirement certificates will be provided. RECs are procured in amount to avoid equivalently grid-based emissions from the production run. For example, if 1 REC results in 100 kWh of emissions reduction at predetermined energy intensity that is provided by the supplier. For electricity consumed, per production batch, RECs are applied to reduce equivalent grid-based emissions. For example, if one HTL production run utilized 500 kWh of electricity, up to five REC (100kWh x 5) would be utilized.
- If electricity is used at the injection site, emissions are calculated by obtaining the local grid emissions factor. RECs are not applied to these emissions.
- The REC emissions factor value includes the embodied and renewable energy production emissions, which is why the net/adjusted emissions factor is not 0 for the respective REC Emissions Factor.
- REC details are attached as supplementary information.

Transport emissions: Biomass feedstock to HTL Conversion Site

- Mass-Distance-Based method formula selected for calculations.
- MASS is monitored for each feedstock delivery and quantified based on documentation such as the Bill of Lading. Feedstock suppliers provide delivery mass values.
- DISTANCE is monitored for road truck transport. Point A (feedstock) and Point B (HTL) are entered into Google Maps to determine the primary trip distance value.
- As per the Transportation Module Section 5.0, The GLEC mass x distance emissions factor includes the return trip impacts.
- For the relevant production run, Mass x Distance is multiplied by the selected EF for the vehicle type utilized.

Transport emissions: bio-oil to injection site

- Mass-Distance-Based method formula selected for calculations.
- MASS is calculated based on the scale ticket at the injection site.
- DISTANCE is monitored for road truck transport. Point A (HTL) and Point B (Injection) are entered into Google Maps to determine the primary trip distance value.
- As per the Transportation Module Section 5.0, The GLEC mass x distance emissions factor includes the return trip impacts.

- For the relevant production run, Mass x Distance is multiplied by the selected EF for the vehicle type utilized.
- The end result is transport emissions that can be properly allocated to the relevant GHG statements.

Transport emissions: consumable transport to the NULIFE site.

- Mass-Distance-Based method formula selected for calculations.
- MASS is calculated based on the supplier invoice.
- DISTANCE is monitored for road truck transport. Point A (consumable location) and Point B (NULIFE) are entered into Google Maps to determine the primary trip distance value.
- As per the Transportation Module Section 5.0, The GLEC mass x distance emissions factor includes the return trip impacts.
- For the relevant production run, Mass x Distance is multiplied by the selected EF for the vehicle type utilized.
- The end result is transport emissions that can be properly allocated to the relevant GHG statements.
- Note: These emissions were considered for the Net negativity example on Certify. They are excluded for regular Removal quantification, as per the materiality assessment.

Transport emissions: HTL wastewater to disposal location.

- Mass-Distance-Based method formula selected for calculations.
- MASS is calculated based on the scale ticket at the injection site.
- DISTANCE is monitored for road truck transport. Point A (HTL) and Point B (disposal location) are entered into Google Maps to determine the primary trip distance value.
- As per the Transportation Module Section 5.0, The GLEC mass x distance emissions factor includes the return trip impacts.
- For the relevant production run, Mass x Distance is multiplied by the selected EF for the vehicle type utilized.
- The end result is transport emissions that can be properly allocated to the relevant GHG statements.

Transport emissions: Analytical samples transport and misc. staff travel.

- Staff travel associated with sample delivery and misc. Work driving is quantified by using NULIFE's vehicle reimbursement logs for staff travel, which provides employees with compensation based on their transport activity when they use their own vehicles. Compensation is determined based on total kilometers travelled. Travel associated with sample delivery is quantified based on the monthly logs.
- Distance-Emission Factor method formula selected for calculations.
- DISTANCE is recorded based on the travel logs.
- As per the Transportation Module Section 5.0, a complete round trip is calculated.
- Distance is multiplied by the selected EF for the vehicle type utilized (gasoline car).
- The end result is transport emissions that can be properly allocated to the relevant GHG statements.

- Note: These emissions were considered for the Net negativity example on Certify. They are excluded for regular Removal quantification, as per the materiality assessment.

Fuel emissions factor for transportation

- Transport emissions are calculated using the weight-distance based emissions factor for transportation: kgCO2e/kg/tonne emissions factor (EF).
- Emissions factors are used for biomass, bio-oil, effluent water, consumables, lab samples, etc. (when relevant).
- Vehicle type used: sludge feedstock, bio-oil and wastewater: tanker truck.
- Vehicle type used: dry feedstock, consumables: flat bed truck.
- Vehicle type used: sample delivery: car.
- The GLEC mass x distance emissions factor includes the return trip impacts.

Notes on return trip impacts: The GLEC factors include the return trip impacts. Below is taken from page 15 of the GLEC Framework:

The GLEC Framework, in line with ISO 14083, calculates emissions per transport chain. To ensure the consideration of empty operations and the accounting of the related emissions, the use of vehicles in transport chains is based on a round trip approach, both for calculation of emission intensities and for the allocation of emissions to consignments in shared transport. Therefore, the necessary return of a vehicle is included, even though freight is usually moved from consignor to consignee in one direction only. This ensures that all emissions related to a transport operation are included.

When using the EF, the distance used in the calculation is based on one direction only, and the empty return trip impacts are included.

Fuel Emissions factor for bio-oil injection

- Currently there is no documented fuel usage for the injection site.
- The injection pumping and brine pumping is electric powered.

Electricity usage for injection site.

- Pumps used: bio-oil injection pump and brine pump
- Pump details provided by the Injection Partner. Details provided include pump horsepower and number of hours of operation to inject the relevant batch of material. The details are used to convert energy usage into kWh.
- Currently the pumping data is communicated to NULIFE and NULIFE manually calculates the associated emissions.
- Applied per injection batch, based on mass injected.
    - A relevant emissions factor is used for SK grid emissions on a mass of CO2e per kWh of electricity basis.

Electricity Emissions Factor

- Regional grid electricity emissions factor is used on a mass of CO_2e per kWh of electricity used basis.

- En84-294-2024-eng.pdf

Embodied Emissions

- As per Section 3.3 of the Embodied Emissions Accounting Module, the mass based method has been selected: Overall mass of equipment, separated by material type, and material emissions factors.
- As per Section 3.2 of the Embodied Emissions Accounting Module, emissions are allocated over the project lifetime.
- As per Section 3.2 of the Embodied Emissions Accounting Module, storage infrastructure is shared amongst multiple entities thus Project Proponent must allocate a percentage of the embodied emissions of that storage infrastructure proportional to the mass of stored material to the total storage capacity of that infrastructure.
- For the HTL conversion site, the mass of steel, stainless steel and polyethylene plastic are determined based on equipment lists.
- For the injection site, the mass of equipment is determined based on equipment lists.
- Allocation of embodied emissions to removals will be reviewed at each Crediting Period renewal and any additional embodied emissions over the original expectation must be accounted for.

Embodied Emissions from HTL conversion site consumables - Municipal Washing Water

- Municipal washing water is used during start up and cooldown cycles of the HTL operations. Quantity of water used is measured and allocated on a per HTL production run basis. The volume is determined by taking the flow rate (L/min) of the trial and multiplying it by the amount of time during operation where water was used as the process inlet material.
- As per section 7.1 of the protocol, the building has out of scope usage such as research & development and corporate administration activities, thus the usage of monthly utility statements for water usage is not used.
- A relevant emissions factor is used for process water emissions on a mass of CO_2e per mass of process water basis.
- Note: These emissions were considered for the Net negativity example on Certify. They are excluded for regular Removal quantification, as per the materiality assessment.

Embodied Emissions from HTL conversion site consumables - Hydrogen Peroxide

- Hydrogen peroxide is infrequently used for cleaning purposes. When used, volume measurements are documented and recorded. The emissions are currently allocated on a per HTL production run basis. A Clean In Place (CIP) solution is prepared in a tote at ~1 wt% chemical/99% water. The volume used, per production batch, is allocated by determining the composition of the solution and how many Liters of CIP solution is used before a given production run.
- A relevant emissions factor is used for hydrogen peroxide emissions on a mass of CO_2e per mass of hydrogen peroxide basis.

- Note: These emissions were considered for the Net negativity example on Certify. They are excluded for regular Removal quantification, as per the materiality assessment.

Embodied Emissions from HTL conversion site consumables - Sodium Hydroxide

- Sodium hydroxide is infrequently used for cleaning purposes. A Clean In Place (CIP) solution is prepared in a tote at ~1 wt% chemical/99% water. When used, mass measurements are taken and allocated on a per production run basis. The volume used, per production batch, is allocated by determining the composition of the solution and how many Liters of CIP solution is used before a given production run.
- A relevant emissions factor is used for sodium hydroxide emissions on a mass of CO_2e per mass of sodium hydroxide basis.
- Note: These emissions were considered for the Net negativity example on Certify. They are excluded for regular Removal quantification, as per the materiality assessment.

Embodied Emissions from HTL conversion site consumables - Activated Carbon

- Coconut shell based activated carbon is used in canisters for gas scrubbing purposes. The mass of activated carbon is determined by supplier information. The replacement cycle is documented. Bio-oil production forecasts are developed and the result is a mass of Activated carbon that is applied based on the mass of bio-oil produced per HTL production batch.
- A relevant emissions factor is used for activated carbon emissions on a mass of CO_2e per mass of activated carbon basis.
- Note: These emissions were considered for the Net negativity example on Certify. They are excluded for regular Removal quantification, as per the materiality assessment.

HTL effluent wastewater emissions - Relevant to CO_2e Misc.

- Direct emissions from the HTL process may occur when wastewater is disposed of. One of the following options are applicable for effluent water treatment:
    - Recycled through the HTL process to minimize effluent water disposal volumes.
    - Disposal down the drain into the municipal sewer system.
    - Treatment onsite using relevant technology and disposed into the municipal sewer system.
    - Transported to a third-party disposal site such as a wastewater treatment plant, injection well, salt cavern or other applicable disposal options.
- CO_2e Misc. may consider, when relevant:
- The CO_2e implications of the wastewater treatment. The formula is:
- Wastewater Treatment Emissions = Water Disposed x EF
    - Water Disposed = Quantity of water disposed, per delivery event. Calculated using disposal ticket or transport bill of lading with disposal volume recording.
    - EF = Relevant Emissions Factor, in $kgCO_2e/L$

- Wastewater Treatment Emissions are calculated in $kgCO_2e/L$, which is then reflected per disposal event, in tonnes of CO_2e.
    - A relevant emissions factor is used on a mass of CO_2e per mass of waste water basis.

- Note: These emissions were considered for the Net negativity example on Certify. They are excluded for regular Removal quantification, as per the materiality assessment.

HTL solid waste emissions - Relevant to CO_2e Misc.

- Solid waste is considered out of scope. The HTL process generates wet waste from the process, which is considered.
- The only solid waste from our company is material that is thrown into a 4 yard garbage dumpster that accumulates solid waste related to research and development and corporate activities. Because the waste is from these activities, it is considered out of scope as per section 7.1 of the protocol.

<u>Monitoring Plan: Bio-Oil/Biomass Storage in Salt Caverns module</u>

An expectations document is provided to the injection partner outlining the monitoring plan requirements pertaining to the required testing for brine and gas as well as reversal notification.

<u>Pre-injection (baseline) brine testing.</u>
The following testing is done prior to NULIFE's first injection batch of bio-oil.

Brine pH
The Injection site has historical brine pH data, which is tested monthly.

- Who is testing? Injection Partner
- Frequency? Historic monthly data is accessible, which is used for the baseline.
- How is testing done? EPP444 method by ISO17025 certified Lab in Saskatoon SK.
- When is testing done? Prior to NULIFE's first injection batch.
- How are the results recorded? Lab analysis sent to the injection site, which is sent to NULIFE. NULIFE stores the files on OneDrive.
- Permit pH requirement: 2.0 – 12.5.

Brine Conductivity/Chloride Content
The Injection site has historical brine chloride data, which is tested monthly.

- Who is testing? Injection Partner
- Frequency? Historic monthly data is accessible, which is used for the baseline.
- How is testing done? Extractable Organic halogens (as chlorides) Method: EOX.A by ISO17025 certified Lab in Saskatoon SK.
- When is testing done? Prior to NULIFE's first injection batch.
- How are the results recorded? Lab analysis sent to the injection site, which is sent to NULIFE. NULIFE stores the files on OneDrive.
- Permit chloride requirement: ≤ 100 mg/L.

Brine carbon wt %

Brine carbon wt% is not currently a required test under the permitting requirements.

- Who is testing? NULIFE
- Frequency? Once
- How is testing done? Carbon wt %
- When is testing done? Prior to NULIFE's first injection batch.
- NULIFE will collect a brine sample and submit samples for Elemental analysis (carbon wt%) at the University of Saskatchewan.
- How are the results recorded? The results will be submitted to NULIFE and NULIFE recorded them on OneDrive.
- Permit requirement: Not required.

Brine Temperature

Brine temperature is not currently a required test under the permitting requirements. Brine temperature is recorded as part of the analysis package provided by the injection partner.

<u>Pre-injection (Baseline) Gas Testing</u>

A sample port has been piped into the brine tank where head space gas can be captured into a tedlar bag.

- Baseline tedlar bag gas sample taken at discharge brine tank prior to 1st injection.
- Baseline GC analysis to confirm sample wt % of CO_2, CH_4 and VOCs.
- Who is testing? The injection partner captures two tedlar bag samples.
- Frequency? At least once.
- How is testing done? Gas sample to be collected from the brine tank by the injection partner using a tedlar bag for next day GC testing. GC testing to be done at the University of Saskatchewan.
- When is testing done? Prior to NULIFE's first injection batch.
- How are the results recorded? Analytical report from the gas analysis, which is delivered to NULIFE and stored on OneDrive.

<u>Injectant and Operational Monitoring</u>

Injectant is defined as the delivery of NULIFE bio-oil. Upon delivery of NULIFE material, the Injection Partner must sample from the truck before offloading. As per permitting requirements, the Injection Partner must store the sample for at least 6 months.The Injection Partner is not required to test each delivery of material, unless there are identified abnormalities that occur during regular operation. At the discretion of the injection partner, they may decide to test the bio-oil for pH, chloride content, lead and/or mercury content.

The Injection Partner has assessed the analytical characteristics of NULIFE's bio-oil since 2022. The characteristics of the bio-oil have been confirmed to be acceptable through analytical testing compared to the permitting requirements for pH, chloride, mercury and lead.

Injectant pH

Isometric requirements are beyond the site permitting requirements for injectant pH, thus the following plan will be implemented:

- Who is testing? NULIFE
- How is testing done? pH meter or strip
- Frequency? Per injection batch
- When is testing done? Sampled during truck loading procedure at NULIFE site.
- How are the results recorded? Recorded by NULIFE and stored on OneDrive.
- Permit pH requirement: 2.0 – 12.5.

Injectant Conductivity/Chloride Content
Isometric requirements are beyond the site permitting requirements for injectant chlorides, thus the following plan will be implemented:

- Who is testing? NULIFE
- Frequency? Per injection batch
- How is testing done? Extractable Organic halogens (as chlorides) Method: EOX.A by ISO17025 certified Lab. Alternatively Total Organic Chlorides (TOX) can be used. TOX result is the sum of organic chlorine, bromine and iodine expressed as chlorine. Method: Rad-113 by an ISO 17025 certified lab.
- When is testing done? Sampled during truck loading procedure at NULIFE site.
- How are the results recorded? Recorded by NULIFE and stored on OneDrive.
- Permit requirement: ≤100 mg/L.

Once consistency of organic chloride results has been established, the frequency of analysis will be reduced. Presently, the injection partner has not requested any more analysis in this area because they are confident that the bio-oil falls within acceptable range for permitting requirements.

$\delta 13$Carbon signature
$\delta 13$Carbon signature testing is useful but not required under the protocol.

- Who is testing? University of Waterloo Isotope Lab.
- Frequency? One sample per feedstock type is the stated requirement in the protocol.
- How is testing done? $\delta 13$Carbon Isotope testing.
- When is testing done? NULIFE is working to establish baseline $\delta 13$Carbon signature results for before and after the HTL process (biomass and bio-oil). We will continue to accumulate this data beyond the requirements of the protocol.
- How are the results recorded? Report provided by the third party lab and stored by NULIFE on OneDrive.

Injectant Temperature
Injectant temperature is not currently a required test under the permitting requirements. NULIFE records a temperature reading for each bio-oil shipment.

<u>Displaced Brine</u>
Displaced brine from the cavern needs to be routinely tested. The injection site must test brine on a monthly basis for pH, chloride, lead and mercury.

Displaced Brine wt% Carbon

Brine Carbon wt%is not a requirement under the site permitting, thus the following plan will be implemented:

- Who is testing? NULIFE
- Frequency? Measurement shall occur every two weeks for 3 months, if the measurements are consistent this can be changed to every month for 6 months, then quarterly for 2 years, and finally every 6 months if these are still consistent.
- How is testing done? A brine sample is collected and mailed to NULIFE in Saskatoon. The sample is tested for Elemental analysis (carbon wt%) testing at either the University of Saskatchewan or a third party lab.
- How are the results recorded? Results are submitted to NULIFE and are then stored by NULIFE on OneDrive.

Displaced Brine pH

- Who is testing? The Injection partner.
- Frequency? Monthly, as per permitting requirements.
- How is testing done? pH meter/strip or EPP444 method by ISO17025 certified Lab.
- When is testing done? The injection partners monthly permitting requirement will be used.
- How are the results recorded? Analytical results are submitted to by the Lab and provided to NULIFE, which NULIFE then stores on OneDrive.
- Permit pH requirement: 2.0 - 12.5.

Displaced Brine Conductivity/Chloride Content

- Who is testing? The injection partner.
- Frequency? Monthly, as per permitting requirements.
- How is testing done? Extractable Organic halogens (as chlorides) Method: EOX.A by ISO17025 certified Lab in Saskatoon SK.
- When is testing done? The injection partners monthly permitting requirement will be used.
- How are the results recorded? Analytical results are submitted to by the Lab and provided to NULIFE, which NULIFE then stores on OneDrive.
- Permit chloride requirement: ≤ 100mg/L.

Displaced Brine Temperature

Brine temperature is not currently a required test under the permitting requirements.

<u>Ongoing Gas Testing</u>

A fixed gas monitor testing for hydrocarbon gases and CO_2 will eventually be placed at the brine tank at the same location as the baseline gas testing location. If carbon bearing gas is detected above 0.01vol%, it could indicate biogas formation and will be investigated.

If the fixed monitor is triggered and it is determined that the alarm isn't from a pre-existing gas release from the cavern, the next step would be:

Tedlar bag sample taken at the same sample location as baseline. GC testing to determine wt% CO_2, CH_4 and VOCs.

- Who is testing? Injection partner samples, NULIFE oversees GC testing.

- Frequency? Once
- How is testing done? Gas sample to be collected from the brine tank by the injection partner using two tedlar bags. Samples to be mailed to NULIFE for next day GC testing.
- When is testing done? After a fixed gas monitor alarm.
- How are the results recorded? Analytical report from the gas analysis, which is delivered to NULIFE and stored on OneDrive.

If carbon bearing gases are detected at levels above the baseline, a reversal could be quantified as per the attribution of reversals section.

Alternatively, NULIFE could advance to IRMS testing to attempt to prove that the gases detected do not belong to NULIFE.

IRMS Stable isotope compositions of carbon containing species.
IRMS could be compared to NULIFE S13Carbon signatures to correlate biogas to NULIFE bio-oil to confirm or void the reversal.

- Who is testing? Injection partner samples, NULIFE oversees IRMS.
- Frequency? Once
- How is testing done? Gas sample to be collected from the brine tank by the injection partner using two tedlar bags. Sample to be mailed to NULIFE for IRMS testing at a third party lab.
- When is testing done? After a fixed gas monitor alarm and GC testing.
- How are the results recorded? Analytical report from the gas analysis, which is delivered to NULIFE and stored on OneDrive.

Note: NULIFE is still investigating the appropriate gas monitor and we anticipate implementation for Q4 2025. In the near term, NULIFE will utilize the brine tank gas testing port to capture tedlar gas bag samples for GC analysis to continuously quantify the wt% of CO2, CH4 and VOCs of capture air samples.

The frequency of tedlar gas testing will be in line with the brine carbon wt% testing. We will capture a gas sample every two weeks for 3 months. If the measurements are consistent this can be changed to every month for 6 months until the gas monitor is installed.

Cavern Integrity

As the Well Operator, the Injection Partner is responsible for monitoring all of the relevant parameters for compliance with their [redacted] permitting ([redacted] Disposal and Injection Well Requirements).

As per the Directive, the required code to follow is CSA Z341.2 Salt Cavern Storage.
CSA Z341 SERIES:22 | Product | CSA Group
The code includes testing associated with

- Material requirements,
- Well completion and conversion,
- Location of underground storage facility,

- Design and development,
- Development and construction,
- Surface facilities (w/ emergency shutdown valves),
- Operations and maintenance,
- Monitoring and measurement,
- Safety,
- Closure: Plugging, abandonment and site restoration.

As per permitting requirements, documentation is primarily stored using The Integrated Resource Information System (IRIS). IRIS is an online business system that supports the development and regulation of [redacted] industry. Through IRIS, the oil and gas industry regularly performs business activities and regulatory tasks with the province online.

Internal Well Integrity

- Required by permitting? Yes
- Measurement method? Pressure testing, scan casing testing to determine how thick the walls are. All testing is done in accordance with permitting requirements.
- Monitoring frequency? Every 5 years
- How is evidence stored? Cavern site documentation.

External Well Integrity

All testing is done in accordance with permitting requirements. Externals of the well are potash/salt. There is no way of knowing the outside integrity. The site contains 5 caverns in close proximity, if there were external well integrity issues, the cavern would lose pressure and seep material into the other caverns, which is not the case (see CSA image below).

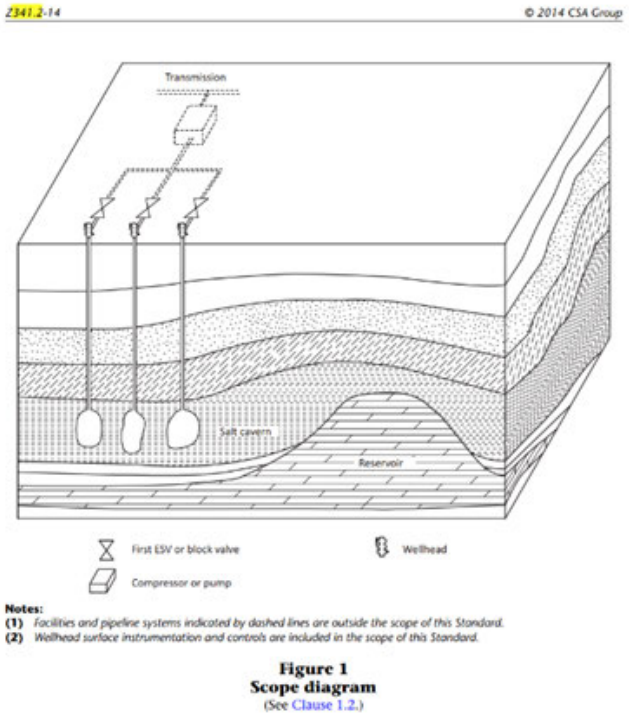


Figure 1
Scope diagram
(See Clause 1.2.)

The testing required by permitting is Mechanical Integrity Testing (MIT).
See attachment CSA Z341 Annex C (informative) Guidelines for mechanical integrity testing of storage caverns. The guidelines cover the following areas:

- General: Caverns should be tested before they are placed in operation to verify their integrity and their capability to contain the stored hydrocarbons. The test pressure at the casing seat should not exceed the design maximum allowable pressure permitted by this Standard.
- Frequency of testing: In addition to testing of caverns prior to the initial storage, this Standard requires periodic retesting. Retesting should be conducted whenever operational or other data, such as unexplained product losses or high pressure in the brine string, indicate that a condition might exist that could adversely affect cavern integrity.
- Nitrogen/brine interface test: more information is provided in annex C.
- MIT data: more information is provided in annex C.
- MIT procedure: more information is provided in annex C.
- Interpreting results: more information is provided in annex C.

MIT results/documentation is maintained by the injection partner.

Corrosion testing - As per CSA Z341 corrosion testing is performed as a part of the operations and maintenance requirements.
Section 10.3 highlights corrosion control including:
10.3.1 Cathodic protection Corrosion control monitoring shall conform to NACE SP0186 for all cathodically protected storage wells. The operator shall
(a) conduct periodic tests to determine whether cathodic protection systems are operating effectively and design current levels are being maintained;
(b) conduct semi-annual on-site inspections and tests of all impressed-current sources for proper operation and correct current output;
(c) inspect all insulating fittings and resistance bonds once per year for proper operation; and (d) consider
(i) measuring casing-to-surface reference electrode potential; and
(ii) conducting casing potential logs.
10.3.2 Interference Operators shall investigate any stray current interference detected by inspections.
10.3.3 Monitoring Fabricated piping shall be inspected annually for corrosion and erosion.
10.3.4 Records Operators shall maintain cathodic protection monitoring records for a period of 15 years after the decommissioning of the storage facility.

Cavern Temperature
Not required by CSA Z341.2. The injection site does not have to test cavern temperature. Based on the temperature of the discharge brine, the injection partner can predict the temperature inside the cavern. Nulife can report the temperature monitoring estimation used by the injection partner.

Cavern Pressure
Testing is required by the permitting body and is done as a part of regular operation. The injection partner logs the cavern pressure daily. To determine cavern pressure, the

pressure is measured at the surface. A one time calculation was done during initial site permitting to determine the cavern pressure. The detailed calculation is utilized to determine cavern based on the surface pressure. See Annex B of CSA Z341.2 calculation of liquid hydrocarbon storage cavern pressure.
Areas covered:

- Typical storage cavern schematic,
- Design pressure calculations in caverns,
- Maximum allowable pressure in casing seat,
- Calculation of pressure in the casing seat.
- How is evidence stored? Cavern site documentation.

Cavern Volume and Fill (Sonar Survey)

- Required by permitting? Yes
- Measurement method? Sonar.
- Monitoring frequency? Annually.
- How is evidence stored? Cavern site documentation.

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Surface Subsidence & Surface Topography

- Required by Permitting? Yes, when the site was initially built. It is not required to be done routinely.
- How is evidence stored? Cavern site documentation.

Underground Source of Drinking Water (USDW) Quality

- Required by permitting? Yes.
- Measurement method? Ground water testing.
- Monitoring frequency? There are no drinking water sources nearby. This testing is done using 8 monitoring wells around the facility that get tested annually for ground water.
- How is evidence stored? Cavern site documentation.
- What is tested for? pH, conductants, total dissolved solids, hardness, alkalinity, various metals, nitrates/nitrites, salts and hydrocarbons.

Salt Permeability
Not required on an ongoing basis by permitting. As per CSA Z341.2.2 salt permeability is required as a part of the well completion process. Salt permeability would have been completed during the well completion process, which was before 1960, for the first license for the facility.

The submitted injection site documents, alongside the nearly 50 year history of cavern usage, is evidence of the salt cavern impermeability.

Porosity and Permeability of caprock and any interbeds
The injection site partner has stated that this is not relevant for the cavern zone utilized.

Existing subsurface features such as faults
Was assessed as a part of the cavern commissioning process. Not an ongoing requirement.

- How is evidence stored? Cavern site documentation.

Reversals

In the event that a gaseous release triggers a gas monitoring alarm, the Injection Partner has confirmed that they will contact NULIFE on the same day as the event. The Protocol states that the VVB should be notified but it is more appropriate for NULIFE to be notified since investigation must take place to determine if the alarms are indicative of a true reversal event or a false alarm.

Any identifiable reversal will be promptly reported to Isometric. As per the Isometric Standard (5.6.1), a report will be made within 1-3 business days from the identification of the potential reversal event. At the verification following a Reversal, NULIFE will report relevant monitoring data for any Reversal which has occurred, to be assessed by the verifier as part of that verification.

B.10 Data collection and storage

Please describe the data collection and storage approach taken, including:

- *How data is transmitted and collected*
- *How data is stored*
- *Length of time for which records are archived*
- *Backup procedures and strategies for identifying and coping with lost or poor-quality data.*
- *Person(s) / organization(s) responsible for measurement and data collection*

How is data transmitted and collected?

Each HTL production run encompasses various data points, which are recorded and collected. The relevant data from each HTL production run is manually entered into a primary Ontario Avenue CDR Excel spreadsheet. The relevant data is then transferred into Isometric's Certify platform to generate “removal data” for GHG statements.

CDR production quantification is done on Isometric’s Certify Platform.

NULIFE’s HTL Production Runs are labelled by their production run date (year-month-date). Example of naming system: 2024-09-04.

Emissions Factor (EF)

- All EFs utilized are uploaded and saved to Certify where they are used as an “input in the relevant “removal data” to generate GHG Statements.

Project Emissions

Embodied Emissions - Equipment for HTL site

- Currently this section has four inputs: HTL Site materials - end of Life, HTL site polyethylene plastic, HTL site stainless steel and HTL site steel.
- Mass of each material is quantified based on equipment lists.

- The data is stored directly in the CDR Excel Spreadsheet under tab: Process Information.
- The relevant Input is labelled “Mass.”
- Each of the four sections has a mass input.
- The unit of measure is kg.
- The total emissions are quantified and must be amortized over the course of the project.
- Embodied input is utilized for GHG statements in Certify under the LCA tab.
- Certify Uploaded Source: Equipment list with mass values.

<u>Sequestrations - Bio-oil storage</u>

Bio-oil - Wt % of Carbon

- Samples are submitted for bi-weekly elemental analysis testing at the University of Saskatchewan Chemical Engineering Lab.
- Carbon wt % of bio-oil results are delivered via raw equipment data using Microsoft Excel. Data is uploaded to OneDrive through an external sharing folder between NULIFE and the lab. The data is then copied into a master excel spreadsheet on OneDrive where the average carbon wt% of the samples is determined.
- The Carbon data is also entered into the primary Ontario Avenue CDR Excel spreadsheet.
- Samples are labelled by their production run date (year-month-date), sample matrix and sample #.
    - Example of naming system: 2024-09-04 bio-oil 1.
- Carbon wt% Input is utilized for GHG Statements in Certify to determine “bio-oil storage” for the relevant “removal data”.
- The relevant Input is labelled “carbon content of product.”
- Certify Uploaded Source: Excel file or screenshot of the Carbon wt% data including the calculated average that is used for this input.

Bio-oil - Mass of product

- Liters of bio-oil production are measured and recorded as a cubic meter value in the Production Run Summary Excel spreadsheet.
- Moisture content of bio-oil data is calculated and stored in the Product Weighing Data Sheet and is recorded in the primary Ontario Avenue CDR Excel spreadsheet.
- Bio-oil density data is stored in the Density Log in the Trial Observations Excel Spreadsheet.
- Using the density of the bio-oil, the cubic meter value (moisture deducted) is converted into kilograms and recorded/stored in the primary Ontario Avenue excel spreadsheet.
- Mass Input is utilized for GHG Statements in Certify determine “bio-oil storage” for the relevant “removal data”.
- The relevant Input is labelled “mass of product”.
- Certify Uploaded Source: Excel file or screenshot of the Trial Report

<u>Removal Activities - Bio-oil Production</u>

Wastewater treatment emissions

- The transport company provides a bill of lading invoice with pick up volume.
- Volume data is transferred into the CDR Excel Spreadsheet under tab: WW.
- Volume input is utilized for GHG statements in Certify to determine “wastewater treatment” for the relevant “removal data”.
- The relevant input is labelled “volume”.
- Certify Uploaded Source: Excel file or screenshot of the Trial Report

Electricity usage power meter

- Power Meter logs are recorded on the Programmable Logic Controller (PLC). The unit of measure is kWh.
- Electricity usage is recorded as a kWh value on a per HTL production run basis. The power meter log results in a single kwh value for the relevant trial.
- The data is recorded in the primary Ontario Avenue CDR Excel spreadsheet.
- The kWh value is entered for the input “Grid electricity usage”.
- Since Renewable Electricity Certificates are acquired on a volumetric matching basis, the kWh value is entered a 2nd time for the input “procured power electricity usage”.
- Certify Uploaded Source: Excel file or screenshot of the Trial Report

Tailgas methane emissions - concentration of warming species in emitted gas

- Samples are submitted for on-demand testing using GC analysis at the University of Saskatchewan Chemical Engineering Lab.
- Samples are labelled by their production run date (year-month-date), sample matrix and sample #. Example naming system: 2024-09-04 tailgas 1.
- Tailgas GC data is delivered in raw GC data through an external sharing One Drive folder between NULIFE and the U of S.
- The raw GC data is stored on OneDrive in a PDF form.
- The Raw GC data is then copied into a master Excel Tailgas spreadsheet that calculates the wt% of CH_4.
- The wt% input is utilized for GHG statements in Certify to determine “tailgas methane emissions” for the relevant “removal data”.
- The input “Concentration of warming species in emitted gas” is entered as a “%” unit of measure.
- Certify Uploaded Source: Excel file or screenshot of the calculated GC results.

Tailgas methane emissions - total mass flow of gas

- The mass flow rate of the tailgas is calculated on a per HTL production batch basis by deducting the output quantities of bio-oil, water soluble oil and biochar.
- This data is auto calculated in the PrimaryOntario Avenue CDR Excel Spreadsheet.
- The mass input is utilized for GHG statements in Certify to determine “tailgas methane emissions” for the relevant “removal data”.
- The input “total mass flow of gas” is entered as a “kg” unit of measure.
- Certify Uploaded Source: Excel file or screenshot of the Trial Report

Municipal water consumable

- Embodied Emissions - Consumable usage for biomass conversion
- Data is recorded in the Trial Data Excel Spreadsheet.

- Data is transferred to CDR Excel Spreadsheet under tab: batch Sample Data.
- Volume input is utilized for GHG statements in Certify to determine “municipal water consumable” for the relevant “removal data”.
- The unit of measure is Liters.
- Certify Uploaded Source: Excel file or screenshot of the Trial Report.
- Note: emissions considered for Isometric Certify Net Negativity example but will be removed from regular CDR quantification, as per the Materiality assessment.

Electricity usage equipment manual log

- Power Meter logs are manually recorded as per the description in section B 9 Monitoring Plan. The unit of measure is kWh.
- The manual log results in a single kwh value for the relevant trial.
- The data is recorded in the energy usage tab in the Trial Data Excel Spreadsheet.
- The data is transferred to the CDR Excel spreadsheet.
- The kWh value is entered for the input “Grid electricity usage”.
- Since Renewable Electricity Certificates are acquired on a volumetric matching basis, the kWh value is entered a 2nd time for the input “procured power electricity usage”.
- Certify Uploaded Source: Excel file or screenshot of the Trial Report
- Note: emissions considered for Isometric Certify Net Negativity example but will be removed from regular CDR quantification, as per the Materiality assessment.

Electricity usage analytical equipment

- Analytical equipment manual logs are manually recorded as per the description in section B 9 Monitoring Plan. The unit of measure is kWh.
- The manual log results in a single kwh value for the relevant trial.
- The data is recorded in the energy usage tab in the Trial Data Excel Spreadsheet.
- The data is transferred to the CDR Excel spreadsheet.
- The kWh value is entered for the input “Grid electricity usage”.
- Since Renewable Electricity Certificates are acquired on a volumetric matching basis, the kWh value is entered a 2nd time for the input “procured power electricity usage”.
- Certify Uploaded Source: Excel file or screenshot of the Trial Report.
- Note: emissions considered for Isometric Certify Net Negativity example but will be removed from regular CDR quantification, as per the Materiality assessment.

Hydrogen Peroxide Consumable

- Embodied Emissions - Consumable usage for biomass conversion
- Data is recorded in the Trial Data Excel Spreadsheet.
- Data is transferred to CDR Excel Spreadsheet under tab: Batch Sample Data
- Volume input is utilized for GHG statements in Certify to determine “hydrogen peroxide consumable” for the relevant “removal data”.
- The unit of measure is Liters.
- Certify Uploaded Source: Excel file or screenshot of the Trial Report.
- Note: emissions considered for Isometric Certify Net Negativity example but will be removed from regular CDR quantification, as per the Materiality assessment.

Sodium Hydroxide Consumable

- Embodied Emissions - Consumable usage for biomass conversion.
- Data is recorded in the Trial Data Excel Spreadsheet.
- Data is transferred to CDR Excel Spreadsheet under tab: Batch Sample Data
- Mass input is utilized for GHG statements in Certify to determine “sodium hydroxide consumable” for the relevant “removal data”.
- The unit of measure is kg.
- Certify Uploaded Source: Excel file or screenshot of the Trial Report.
- Note: emissions considered for Isometric Certify Net Negativity example but will be removed from regular CDR quantification, as per the Materiality assessment.

Activated Carbon Consumable

- Embodied Emissions - Consumable usage for biomass conversion.
- Data is recorded in the Trial Data Excel Spreadsheet.
- Data is transferred to CDR Excel Spreadsheet under tab: Batch Sample Data
- Mass input is utilized for GHG statements in Certify to determine “activated carbon consumable” for the relevant “removal data”.
- The unit of measure is kg.
- Certify Uploaded Source: Excel file or screenshot of the Trial Report.
- Note: emissions considered for Isometric Certify Net Negativity example but will be removed from regular CDR quantification, as per the Materiality assessment.

Sodium Hydroxide, Hydrogen Peroxide and Activated Carbon usage is tracked using our Consumable Tracking Excel Sheet.

Wastewater Transport emissions (tanker) - Distance Travelled
Distance of effluent wastewater transported from conversion site to disposal site

- Distance is determined via Google Maps.
- Distance data is transferred into the Excel File: Incoming Outgoing Delivery Log Ontario Avenue .
- Distance input is utilized for GHG statements in Certify to determine “wastewater transport emissions” for the relevant “removal data”.
- The unit of measure is km.
- Certify Uploaded Source: Mapping screenshot i.e. Google Maps or transport documentation.

Wastewater Transport emissions (tanker) - Mass of Load
Mass of effluent wastewater transported from conversion site to disposal site.

- The transport company provides a bill of lading invoice with pick up mass.
- Mass data is transferred into Excel File: Incoming Outgoing Delivery Log Ontario Avenue .
- Mass input is utilized for GHG statements in Certify to determine “wastewater transport emissions” for the relevant “removal data”.
- The unit of measure is kg.
- Certify Uploaded Source: Transport Bill of Lading, invoice or other documentation.

Activated Carbon Transport emissions (flat deck) - Distance traveled

Distance of consumable transported from consumable company location to conversion site.

- Distance is determined by using Google Maps.
- The data is uploaded directly to the CDR Excel spreadsheet.
- Distance input is utilized for GHG statements in Certify to determine “activated carbon transport emissions” for the relevant “removal data”.
- The unit of measure is km.
- Certify Uploaded Source: Mapping screenshot i.e. Google Maps or transport documentation.
- Note: emissions considered for Isometric Certify Net Negativity example but will be removed from regular CDR quantification, as per the Materiality assessment.

Activated Carbon Transport emissions (flat deck) - Mass of load
Mass of consumables transported from supplier to conversion site

- The invoice from the consumable supplier contains a mass value for the purchased material.
- NULIFE uploads the document to OneDrive.
- The data is transferred to the CDR Excel spreadsheet.
- Mass input is utilized for GHG statements in Certify to determine “activated carbon transport emissions” for the relevant “removal data”.
- The unit of measure is kg.
- Certify Uploaded Source: Transport Bill of Lading, invoice or other documentation.
- Note: emissions considered for Isometric Certify Net Negativity example but will be removed from regular CDR quantification, as per the Materiality assessment.

Hydrogen peroxide/sodium hydroxide Transport emissions (flat deck) - Distance traveled
Distance of consumable transported from consumable company location to conversion site.

- Distance is determined by using Google Maps.
- The data is uploaded directly to the CDR Excel spreadsheet.
- Distance input is utilized for GHG statements in Certify to determine “Hp/Sh transport emissions” for the relevant “removal data”.
- The unit of measure is km.
- Certify Uploaded Source: Mapping screenshot i.e. Google Maps or transport documentation.
- Note: emissions considered for Isometric Certify Net Negativity example but will be removed from regular CDR quantification, as per the Materiality assessment.

Hydrogen peroxide/sodium hydroxide Transport emissions (flat deck) - Mass of load
Mass of consumables transported from supplier to conversion site

- The invoice from the consumable supplier contains a mass value for the purchased material.
- NULIFE uploads the document to OneDrive.
- The data is transferred to the CDR Excel spreadsheet.
- Mass input is utilized for GHG statements in Certify to determine “Hp/Sh transport emissions” for the relevant “removal data”.
- The unit of measure is kg.

- Certify Uploaded Source: Transport Bill of Lading, invoice or other documentation.
- Note: emissions considered for Isometric Certify Net Negativity example but will be removed from regular CDR quantification, as per the Materiality assessment.

Removal Activities - Bio-oil Storage

Equipment Embodied Emissions per MT of injectant

- Injection site equipment embodied emissions.
- An emissions factor to determine Mass based CI emissions is attached in Certify.
- The Mass is determined based on the invoice from the injection site and appropriate documentation such as the weight scale ticket.
- NULIFE uploads the document to OneDrive.
- Mass input is utilized for GHG statements in Certify to determine “equipment embodied emissions per MT of injectant” for the relevant “removal data”.
- The unit of measure is kg.
- Certify uploaded source: Bio-oil injectant mass from Bill of lading, injection site invoice or other documentation.

Injection electricity usage

- Injection site electricity usage.
- The unit of measure is kWh.
- The Injection Partner has provided specifications on their primary pump and its horsepower along with frequency the pump is running for the relevant injection batch.
- The relevant data is transferred to the Isometric Certify Calculations Excel Spreadsheet, which converts the data to kWh and calculates tonnes of CO_2e per injection event using the relevant emissions factor.
- kWh input is utilized for GHG statements in Certify to determine “injection electricity usage” for the relevant “removal data”.
- Certify uploaded source: Calculated data quantifying the kWh value via excel file or screenshot.

Removal Activities - Bio-oil Transport

Bio-oil transport to storage site via truck (tanker) - Distance travelled

- Distance of bio-oil transported from conversion site to injection site.
- Distance is determined by using Google Maps.
- The data is uploaded directly to Certify.
- Distance input is utilized for GHG statements in Certify to determine “bio-oil transport to storage site via truck” for the relevant “removal data”.
- The unit of measure is km.
- Certify uploaded source: Mapping screenshot i.e. Google Maps or transport documentation.

Bio-oil transport to storage site via truck (tanker) - Mass of load

Mass of bio-oil transported from conversion site to injection site

- The injection site provides an injection statement including ticket with delivery mass based on arrival and departure weight.

- NULIFE uploads the document to OneDrive.
- The data is uploaded to Certify.
- Mass input is utilized for GHG statements in Certify to determine “bio-oil transport to storage site cia truck” for the relevant “removal data”.
- The unit of measure is metric tonne.
- Certify uploaded source: Bill of lading, invoice or other documentation.

Removal Activities - Biomass sourcing

Biomass (dry/wet) transportation to production site via truck (flat deck/tanker) - Distance travelled

- Distance of biomass transported from supplier to conversion site
- The location of the supplier is recorded on bill of lading or other documentation.
- Distance is determined by using Google Maps.
- The data is uploaded directly to Certify.
- Distance input is utilized for GHG statements in Certify to determine “biomass transport to storage site via truck” for the relevant “removal data”.
- The unit of measure is km.
- Certify uploaded source: Mapping screenshot i.e. Google Maps or transport documentation.
- The allocation approach may be used to separate biomass transport emissions between removal batches.

Biomass (dry/wet) transportation to production site via truck (flat deck/tanker) - Mass of load

- Mass of biomass transported from supplier to conversion site
- The transport company provides a bill of lading with delivery mass provided by the feedstock supplier.
- NULIFE uploads the document to OneDrive.
- The data is transferred to Certify.
- Mass input is utilized for GHG statements in Certify to determine “biomass transport to storage site via truck” for the relevant “removal data”.
- The unit of measure is kg.
- Certify uploaded source: Bill of lading, invoice or other documentation.
- The allocation approach may be used to separate biomass transport emissions between removal batches.

Removal activities - MRV

Staff travel sample delivery (gasoline car) - Distance travelled

- Transport events are documented in employee monthly driving logs, which includes transport distance per event, in Km.
- These documents are uploaded by employees to the Dext payment software where employees are reimbursed via monthly expense reports.
- Distance input is utilized for GHG statements in Certify to determine “staff travel sample delivery” for the relevant “removal data”.
- The unit of measure is km.
- Certify uploaded source: Employee travel reimbursement report + mapping software screenshot i.e. Google Maps.

- Note: emissions considered for Isometric Certify Net Negativity example but will be removed from regular CDR quantification, as per the Materiality assessment.

Removal activities - MRV
Operator travel to work with return trip - Distance travelled

- Based on the location of residence the distance is determined with the end point being the Nulife plant. A return trip home is assumed.
- Distance input is utilized for GHG statements in Certify to determine “staff travel operators to site” for the relevant “removal data”.
- The unit of measure is km.
- Certify uploaded source: Distance calculations + mapping software screenshot i.e. Google Maps.
- Note: emissions considered for Isometric Certify Net Negativity example but will be removed from regular CDR quantification, as per the Materiality assessment.

<u>Others</u>
Biomass Feedstock Counterfactual Data

- Data related to counterfactual calculations is uploaded directly into the CDR Excel Spreadsheet under tab: Feedstock, Counterfactual.

Biomass conversion batch operation time

- The trial start time, end time and total operation time are recorded in the Trial Observations Excel spreadsheet.
- The data is transferred to the CDR Excel spreadsheet.

Mass of fuel used in biomass conversion

- Currently the HTL system does not utilize fuel to convert biomass into bio-oil.
- When relevant, the fuel data will be recorded in the Trial Observations spreadsheet and subsequently transferred to the CDR Excel spreadsheet.

Any information related to feedstock or injection partners is sent to NULIFE upon request or via agreed upon arrangements. NULIFE then stores the documentation internally.

How is data stored?
NULIFE uses Microsoft OneDrive to store internal and external data. OneDrive backs up files that are synced to it. OneDrive is a cloud storage service that allows users to back up their files and folders to the cloud. Once backed up, files are accessible from any device.

All raw data is saved and stored on NULIFE’s One Drive. This raw data is also stored in locations related to their place of origination such as:

- NULIFE’s Programmable Logic Control (PLC) Ignition software,
- Raw Bio-oil carbon wt% and tailgas GC analytical reports,
- Bill of lading and invoices from third parties such as the injection site and transportation companies are stored with the respective third parties as well as through NULIFE’s third party financial document management software (Dext).

Length of time for which records are archived?
Once uploaded to OneDrive, records will be stored, at minimum, in accordance with Isometric Standard requirements. Internal data, such as Carbon wt% values and injection site scale tickets are stored for at least 5 years.

Backup procedures and strategies for identifying and coping with lost or poor-quality data?
If any OneDrive files or folders were deleted, overwritten, corrupted, or infected by malware, you can restore your entire OneDrive to a previous time. If you don't want to restore everything, you can also restore deleted files from the OneDrive recycle bin or restore a previous version of a file.

NULIFE's PLC Ignition software is currently regularly manually backed up and we are in the process of implementing an automated back up procedure where the data is regularly backed up to the Cloud (One Drive).

Person(s) / organization(s) responsible for measurement and data collection?
NULIFE GreenTech is responsible for measurement and data collection. It is our responsibility to ensure that our partners are collecting and providing the appropriate data for registry compliance.

B.11 Reversals

Please complete a reversal risk assessment for your Project, including consideration of the guidance in the Risk of Reversal Questionnaire below.

Overall risk of reversal (based on risk reversal questionnaire score; please select one):

0: Very Low Risk
1-2: Low Risk
3-4: Medium Risk
5+: High Risk

[Optional] Additional details/assessment of reversal risk mechanisms:

#	Question	If answered "Yes"	If answered "No"	Response
1	Is a reversal directly observable with a physical or chemical measurement as opposed to a modeled result?	Proceed to questions 2-9	Proceed to questions 8-9	Yes
2	Is the carbon being stored in an impermeable geologic system? (e.g., salt cavern)	Proceed to questions 8-9	Add 1 to Risk Score and	Yes

			proceed to questions 3-9	
3	Is the carbon being stored organic?	Add 1 to Risk Score		~
4	Are conditions for methane production present (anaerobic conditions, lignin content)? storage reservoir has a less than 10,000 year durability?	Add 1 to Risk Score		~
5	Does this approach have a material risk of reversal due to natural disasters including, but not limited to, floods, storms, earthquakes, fires, etc.?	Add 1 to Risk Score		~
6	Does this approach have a material risk of reversal due to human-induced events from outside actors, such as change in farming practices, change in ownership and management of Project sites, or similar?	Add up to 2 to Risk Score		~
7	[Applicable only for subsurface storage] Is the carbon being stored with trapping mechanisms preventing reversals? (e.g., multiple confining layers, CO_2 dissolves or solidifies)	Minus 1 to risk (unless 0)		~
8	Is there 10+ years of monitoring and/or lab data demonstrating low project risk?	Minus 2 to risk (unless 0)		Yes

9	Does this pathway have a documented history of reversals?	Please consider the frequency and severity of previous reversals, and the shared characteristics between documented reversals and the present project. For pathways with no documented history of reversals, add 0 to the Risk Score. For pathways with a history of frequent reversal, add 2 to Risk Score. For pathways with a limited history of reversals, add 1 to Risk Score.		No
10	Is there one or more Project-specific factors that merit a high risk level?	Please consider the number and severity of risks identified. If one low or medium severity risk is identified, add 1 to Risk Score. If multiple risks are identified, or if any high severity risks are identified, add 2 to Risk Score		No
Total Risk Score				0

B.12 Uncertainty assessment	
Specify how uncertainty is considered, and how removals are to be conservatively estimated, in accordance with Section 2.5.7 "Uncertainty" of the Isometric Standard.	
Please specify which option(s) were used in consideration of uncertainty (*one or multiple options*).	
●	Conservative estimate of input parameters
●	**Variance propagation (selected)**
●	Monte Carlo Simulations

Sensitivity analysis uploaded as an attachment titled: Attached.	
●	I have uploaded a sensitivity analysis in accordance with the guidance in the Isometric Standard

Uncertainty Assessment
The Isometric Certify Sensitivity Analysis flagged two items with an impact on net removal of > 20%. These items are mass of product (bio-oil) and Carbon content of product (bio-oil). Both values provide uncertainty justification:

Mass of Product: The datapoint is accurate primary data and does not require an uncertainty discount. Bio-oil mass is measured data, with high certainty. Original values are used without a standard deviation.

Carbon content of product: The datapoint is accurate primary data and does not require an uncertainty discount. Carbon wt% is measured data, with high certainty due to the elemental analysis equipment operational requirements of utilizing standards to confirm accuracy of equipment, the frequency of sampling and duplicate samples being tested. Each HTL Production Run, which is defined as a day of operation, is tested for carbon wt% with three (3) unique samples of bio-oil. Each sample is tested in duplicates in the elemental analysis machine to determine the carbon wt% value. During one HTL production run, August 5, 2025, there is a gap in elemental analysis data. For this trial, the conservative value of 70% carbon is assumed, which is 5% less than the weighted average for the entire HTL production run. It is important to note that the bio-oil mass during the August 5th trial is < 0.5% of the entire bio-oil mass for the injection batch.

The remaining data points do not impact net removals by greater than 1%, thus further analysis is not required.

Conservative Assumptions
Assumptions and choices made in the assessment are conservative.

Nearly all the data points are primary measured data including:
- Bio-oil Mass,
- Bio-oil Carbon wt%,
- Wastewater Mass,
- Tailgas Mass,
- HTL Electricity Usage,
- Biomass Mass.

At the injection site, a 20% increase in electricity usage is assumed due to the lack of primary measured data from the injection site.

Transport event distances are based on factual routes between points A and B. The mass x distance emissions factors include the return trip impacts. Additionally, a 5% increase in transport distance is added via standard deviation to account for potential extension of routes.

Materiality Assessment
Certain SSRs were omitted from this removal due to the Materiality Assessment exercise, as per GHG Accounting V1.0. More details are below:

Title: Materiality Assessment				
Date: September 10, 2025				
Protocol: GHG Accounting V1.0				
Data Used: Isometric's Certify Net Negativity NULIFE Example				
CO2e Removal (tCO2e)	4.188			
If materiality of the SSR is < 1%, then the emissions associated with the SSR may be excluded from the GHG Statement boundary, provided that the total exclusions make up < 1% collectively. If the sum of SSRs that individually are < 1% becomes > 1%, then the SSRs with the highest Materiality must be included in the GHG Statement boundary until the sum of SSRs becomes < 1%. It is not necessary to calculate emissions related to SSRs that are < 1% using more accurate data if the emission calculations for the Materiality assessment were conducted using a high-level approach.				
The estimated total net CO2e Removal should be calculated using actual data and following all required methodologies for SSRs considered to be material.				
SSR	SSR tCO2e	% of CO2e Removal	Included/Excluded?	Notes
Biomass (wet) transportation to production site via truck (tanker)	0.005	0.11938873	Included	Important SSR that we will continue to include
Biomass (dry) transportation to production site via truck (flat deck)		0	Included in future	example did not include dry biomass. Will include SSR when events take place
Wastewater treatment emissions	0.003	0.071633238	exclude	
Clean Power procurement (REC) Power Meter	0.092	2.196752627	Included	HTL Site - Will include SSR when events take place
Tailgas methane emissions	0.03	0.716332378	Included	HTL Site - Will include SSR when events take place
Municipal water consumable	0.001	0.023877746	exclude	
Clean power procurement (REC) equipment manual log	0.005	0.11938873	exclude	
Clean power procurement (REC) usage analytical equipment	0.001	0.023877746	exclude	
Hydrogen Peroxide Consumable	0.009	0.214899713	exclude	
Sodium Hydroxide Consumable	0.001	0.023877746	exclude	
Activated Carbon Consumable	0.003	0.071633238	exclude	
Wastewater Transport emissions (tanker)	0.002	0.047755492	Include	Will include SSR when events take place due to potential for change in transport distance
Activated Carbon Transport emissions (flat deck)	0.001	0.023877746	exclude	
Hydrogen peroxide/sodium hydroxide transport emissions (flat deck)	0.001	0.023877746	exclude	
Bio-oil transportation to storage site via truck (tanker)	0.038	0.907354346	Include	Will include SSR when events take place
Equipment Embodied Emissions per MT of injectant	0.001	0.023877746	Include	Will include SSR when events take place
Injection electricity usage	0.007	0.167144222	Include	Will include SSR when events take place
Staff travel sample delivery (gasoline car)	0.001	0.023877746	exclude	
staff travel operators to site	0.007	0.167144222	exclude	
Sum of SSRs excluded (must be < 1%)	0.208	**0.787965616**		still below the 1% threshold

B.13 Use of Models

Please describe your use of models (if any) for quantification, monitoring, and meeting specified Protocol requirements. Describe the specific model and simulations used, with enough detail so that the work could be replicated.

Please provide model validation results to demonstrate model accuracy, and include an assessment of model uncertainty.

The only use of models for the quantification of this delivery is feedstock/baseline CO2e Decay Rate counterfactual, where a quantification calculation was used to predict the decomposition rate of the biomass carbon over 15 years in their baseline state. This is only used when relevant to the baseline scenario.

B.14 [Applicable to closed systems] Storage site overview

Please describe the storage well / site used, and complete the following information.

This question is only applicable to closed systems. For other pathways, please answer "N/A" or simply delete this question.

The injection facility utilizes underground caverns that were originally designed for high pressure gas storage up to 3,000 psi. This results in caverns that offer superior impermeability, safety, durability and an unmatched environmental integrity. This world-class facility utilizes these sealed underground caverns along with the most demanding environmental standards to ensure the integrity of operations. The salt caverns utilized have tops that are approximately 1,055 meters underground, and their bottoms are around 1,190meters underground. The distance from top to bottom of the caverns is approximately 140 meters.

NULIFE is utilizing the injection site's Cavern #1 for bio-oil storage. Cavern #1 was previously used for waste disposal purposes. It is currently ~66% full. The cavern is currently not in use for typical waste disposal operations. Waste disposal operations are currently directed into cavern #4. Cavern #1 will be utilized solely for NULIFE bio-oil to fill the cavern before it is closed. It is estimated that there is 50,000 m3 of capacity remaining for NULIFE bio-oil.

Currently, Cavern #1 and #4 utilize the same injectant and discharge brine tanks. Cavern #2, #3 and #5 are available, when more storage is needed. Considering existing caverns alone, the total capacity of the site is 500,000m3.

Injection Steps:

1. The injectant truck enters the site through the main gate and drives over the scale to determine an entry weight.
2. The injection partner samples the bio-oil for permitting purposes.
3. The injectant truck drives to the pad, where the truck connects to the site hose. The truck offloaded the bio-oil using the truck pump.
4. The bio-oil is pumped into the sludge tank.
5. The injectant truck drives over the scale to determine the exit weight and leaves the site after receiving the exit scale ticket and other exit documentation.
6. The injection partner utilizes the slurry pump to inject the bio-oil into the relevant cavern.
7. Cavern brine is discharged and enters the brine tank through a closed system.
8. Clean brine is pumped into the onsite brine disposal well.

Monitoring overview (*please summarise the current monitoring in place, as required by the relevant permit and in accordance with the protocol monitoring requirements. You may provide more information on individual monitoring parameters in the Project Monitoring Plan):*
Monitoring is discussed in detail in the B.9 Monitoring Plan section of this PDD.

Storage well/site location (Address / GPS coordinates): The facility is located at

LSD: [redacted]

[redacted]
Regulatory Authority / Jurisdiction: [redacted]
Well Class (if applicable): Salt Cavern.
Permit number. The permitting body operates through facility codes, see below. SIR License: [redacted] Operator Code: [redacted] Cavern #1 Facility Code: [redacted] Cavern #4 Facility Code: [redacted]
Permitting authority: [redacted]
Permit validity start date: 2006
Permit validity end date: Ongoing validity under the [redacted] Facility code. All caverns and disposal wells onsite fall under this umbrella for the facility code. Once a license is in place, it is in place as long as compliance is maintained. Amendments can be submitted via permitting processes to add additional codes under the primary facility code.
Well/site storage capacity (total): Each cavern empty is about 150,000 cubic meters of total capacity on average. The range is between 138,000 to 163,000 cubic meters. There are a total of 5 caverns at this location. Total site remaining capacity is estimated at +500,000 cubic meters.
Well/site storage capacity (used): Cavern 4: 66% to 75% used. Cavern 1: 66% used. Three new caverns: 0% used in all three.
Well/site storage capacity (available): Cavern 4: 25% to 44% available. Cavern 1: 33% available. Three new caverns: 100% available in all three.

Section C – Duration & Crediting Period

C.1 Project timeline
Please indicate the projected start date of your Project and, if applicable, its expected operational closure date.
Start date of Project: August 1, 2025

[Optional] Expected operational closure date of Project: unknown

C.2 Project closure

Please describe the conditions under which the Project will be considered closed, and describe the Project Closure Plan – outlining any post-cessation actions that will be undertaken upon Closure of the Project.

Definition of Project cessation:

1. Once the caverns at the injection site have reached capacity or
2. NULIFE stops injecting at the injection site.

The termination of the project does not terminate the post injection monitoring. This monitoring must continue until the salt cavern has been properly closed.

Closure Plan:
As per the [redacted], the Injection Partner is responsible for the closure and/or ongoing monitoring of the cavern after NULIFE has ceased operations at the facility. The closure plan is in compliance with the Isometric Closure Plan requirements.

As per CSA Z341-2.2 requirements, the cavern closure plan includes:

- Sonar survey:Where more than five years have elapsed since the previous sonar survey, another survey shall be conducted prior to wellbore abandonment to determine the final shape and area of the cavern.
- Mechanical integrity tests: A pressure test shall be performed on the cavern and wellbore prior to abandonment using brine only or gas/brine interface testing. If any significant leakage is detected behind the casing, the operator shall determine the leak location and carry out remedial work to ensure isolation of cavern fluids.
- Well abandonment design shall (a) ensure that the storage zone is isolated completely from all other porous or hydrocarbon-bearing horizons; (b) prevent the migration of hydrocarbons or water from one horizon to another; and (c) ensure that the surface is returned to near-original condition,
- Downhole equipment removal: Downhole equipment and uncemented casing and tubing strings shall be removed from the wellbore prior to commencing wellbore abandonment operations. Casing or other equipment essential to the proper control of a well shall not be removed.
- Abandonment plugs: details on the neat cement utilized for abandonment well plugs can be found in CSA Z341.
- Surface abandonment: Surface abandonment procedures shall include (a) determining the cause of any surface casing vent flow and repairing it prior to commencing surface abandonment; (b) ensuring that no hydrocarbons are migrating from the wellbore; (c) where the surface casing is not cemented to the surface, plugging the production casing with cement from a depth of 200 m to the surface; (d) cutting casing strings 1 m below ground level; and (e) capping the surface casing with a welded steel plate.

- Site restoration: Site restoration procedures shall include: (a) disposing of all liquid waste; (b) clearing the area of debris; (c) draining and filling excavations; (d) removing unused concrete bases, machinery, and materials; (e) levelling and restoring the site to its original condition or as close to its original condition as possible; and (f) conducting periodic site inspections.

C.3 Crediting Period
Please indicate the planned start date and duration of your crediting period. The crediting start date may either be your Project's start date or up to two years prior to design submission, whichever is later. Unless otherwise specified in the relevant Protocol, the maximum crediting period is 5 years.
Start date of crediting period: Aug 1, 2025
Total length of crediting period: 5 years.

D – Environmental and social impacts

D.1 Analysis of environmental and social impacts	
Please provide an assessment of the environmental and social impacts of the Project, in accordance with Section 3.7 "Environmental and Social Impacts" of the Isometric Standard. *For each aspect of the assessment, demonstrate how the risks have been assessed and if applicable, what mitigation plan is in place to prevent them. If some aspects are not applicable to your project, justify how you determined it.* *A full Environmental and/or Social Impact Assessment (EIA and/or SIA) is only required if impacts are considered significant and/or if required by the host jurisdiction.*	
x	[Optional] Due to the low risks of the project, we have not attached an optional full EIA document. An EIA is not required by local regulators. The impacts are disclosed in the relevant section below.
x	[Optional] Due to the low risks of the project, we have not attached an optional full SIA document. An SIA is not required by local regulators. The impacts are disclosed in the relevant section below.
x	I acknowledge responsibility for reporting potential environmental and social impacts identified to Isometric and environmental regulators

D.2 Environmental Impacts	
D.2.1 Resource efficiency and pollution prevention, including pollutant emissions to air, pollutant discharges to water, noise and vibration, generation of waste and release of hazardous materials, chemical pesticides and fertilizers.	
x	Above risks are applicable to this Project
●	Not applicable to this Project

Environmental impacts need to be considered for:

1. HTL Conversion
2. Injection Site

Since we focus on true waste streams and by-products that are already being disposed of, we provide positive environmental impacts by accepting waste biomass. Our company is built around waste reduction. SDG 6.3 and SDG 12.4 - The waste managed by NULIFE would otherwise end up in farmland, lagoons, landfills, or wastewater treatment plants. These options pose their own risks due to the level of contaminants within the waste. Contamination includes acidic pH level, PFAS, micro plastics and pathogens. By sequestering these wastes, NULIFE helps minimize adverse impacts on human health and the environment.

HTL Conversion

NULIFE GreenTech is located at 1725 Ontario Ave, Saskatoon within IH Zoning (Heavy Industrial). The zoning allows for nuisances such as noise, dust, odours, etc. One of our street neighbours is a local asphalt refinery.

NULIFE's company wide safety program include the following environmental policy:

NULIFE GreenTech is committed to minimizing the impact of its activities on, and protecting, the environment. NULIFE GreenTech's environmental commitment will be integrated with corporate decisions and strategies based on the following guiding principles.

Guiding Principles - NULIFE GreenTech is committed to:

- Complying with applicable environmental law, industry standards, and its environmental policies;
- Making environmental considerations an integral part of its planning process and promoting them within the organization;
- Operating its vehicles and facilities in a manner that protects the environment and the safety and health of its employees and the public;
- Promptly providing relevant information to all stakeholders affected by its operations and being responsive and sensitive to legitimate stakeholder concerns;
- Protecting the land and peoples where we operate;

- Promoting environmental sustainability through company established communication methods internally and with stakeholders;
- Identifying and mitigating any potential adverse effects of its operations on the environment and surrounding communities;
- Being prepared for and responding to emergencies in a prompt and efficient manner;
- Ensuring all employees are aware of the requirement to report spills of any quantity; and
- Ensuring that its employees are fully informed of their responsibilities and are trained to protect the environment while performing their duties.

NULIFE GreenTech management, employees, and contractors will be required to adhere to this policy at all times.

Our Saskatoon Plant is already operational. We are considered light manufacturing and only require a business license to operate at our current scale. We have Technical Safety Authority Saskatchewan (TSASK) approval of our HTL processor design, which does not need to be renewed until 2027. We have approval from the Saskatchewan Ministry of Environment (MOE) to operate and a signed letter confirming that our Saskatoon Ontario Avenue project currently does not require permitting.This letter is attached as a support document.

NULIFE has completed a hazardous substance and waste dangerous good characterization assessment for products and substances present at the facility and have concluded that an Saskatchewan MOE Approval to Store Hazardous Substances and/or Waste Dangerous Goods is not required. This has been confirmed by the MOE in the attached documentation..

All testing performed on feedstock and effluent products is completed by trusted third party labs. Wastewater is tested at the Saskatchewan Research Council (SRC). Primary testing of other products is done by Alberta Innovates and the University of Saskatchewan Chemical Engineering Lab. Gas analysis testing is completed using an Agilent Gas Chromatography (GC) system.

<u>Feedstocks</u>
Presently the non-hazardous organics that are being utilized for processing are sludges, oat hulls, and bran waste.

The feedstock storage area of the plant provides a safe and acceptable storage and handling solution for the feedstock for HTL processing. The HTL process requires the organic particles and water be mixed to create a slurry, which will then allow the HTL reaction and process to happen effectively.

There are four 10,000 Liter poly tanks in the building to store wet feedstocks and/or wastewater.

Additionally, there is a sump pit in the floor to capture any spillage. This sump pit is not connected to the City's sewer. It is pumped out by a contractor for safe disposal. The contractor also removes the wastewater for safe disposal.

The dry non-hazardous organic feedstock such as bran waste and oat hulls are delivered on pallets in bulk bags or in consumer packaging. This material is stored in secure and dry areas.

Bio-oil

We have injected commercial loads of our bio-oil and there has been no smell issue at the injection site. The Bio-oil is non hazardous.This can be validated by the attached Safety Data Sheet (SDS) as well as Toxicity Characteristic Leaching Procedure (TCLP) testing which can be provided upon request. Bio-oil is an oxygenated carbon product derived from biomass carbon and is black or brown in color. It has a very high flashpoint temperature 150-200C and autoignition temperature of <450C. The bio-oil also contains high levels of oxygen and moisture, which results in the high flashpoint temperature. Our bio-oil is not classified as hazardous according to the Transportation of Dangerous Good Regulation (Canada).

The bio-oil storage and handling system includes a 400 barrel tank installed within a containment system on the west side of the exterior yard. The secondary containment system has been sized according to Saskatchewan Directive S-01 "Upstream Petroleum Industry Storage Standards". The vent gas off the tanks are piped into an activated carbon canister.

Tailgas - Air

A stainless-steel flash tank receives the effluent from the HTL process. The effluent consists of water, bio-oil, and gases. The tank allows for the collection of the liquid effluent products (water, bio-oil) and separation of the gases. The tank has level control monitoring liquid effluent levels with a control system which automatically stops the process if the tank is not emptied by the operator.

The process gases are vented from the tank and are scrubbed with activated carbon to remove the minor VOC content in the gas stream. The process gas is cooled through a heat exchanger prior to entering the activated carbon. The tailgas from the activated carbon are safely vented to the atmosphere.

The absorbed gas in the activated carbon is trapped forever. The only way you can get it out is through intentional steam treatment. The activated carbon is safely disposed of in a local landfill. From a carbon accounting perspective, we are sampling gas prior to the drum to determine CH_4 levels, so we do not understate CH_4 release. The drum does not absorb CH_4. The drum absorbs VOCs. We do not consider VOCs from a carbon accounting perspective. We are considering a GWP of 0 for CO, so we do not track any emissions from a carbon accounting perspective. Carbon monoxide (CO) has a Global Warming Potential (GWP) of approximately zero over a 100-year period, because it does

not act as a direct greenhouse gas—it doesn't absorb infrared radiation like CO_2 or methane.

As of 2025-03, we are not considered an Industrial Source (air quality) due to the following requirements for MOE air quality permitting:

Industrial Sources - 1-7 (m) a facility that annually emits or is expected to annually emit:

- more than 10 metric tonnes of one type of Class I air contaminant listed in Appendix:
- more than 25 metric tonnes of any combination of three or more types of Class I air contaminants listed in the Appendix.

Class I appendix:

- sulphur dioxides (SO2)
- oxides of nitrogen (expressed as NO2)
- fine particulate matter (PM2.5)
- coarse particulate matter (PM10)
- carbon monoxide (CO)
- volatile organic compounds (VOCs)

Source: Publications Centre

Based on all the testing of the gas stream over the last several years, our CO_2 concentration is typically 98-98.5%, with CO 1-2%, and remaining concentrations being methane and VOCs. The activated carbon is effectively removing VOC emissions to the atmosphere, thereby reducing total emissions.

Detailed produced gas analysis will continue to be conducted, as the information is used to assess the effectiveness of the overall HTL process.

Our process tailgas emissions are safe. We have had third party air consultants assess and model our emissions and determined they are safe and there are no adverse health concerns associated. Our emissions are well below the ambient air quality guidelines.

Health & Safety

The process tailgases have been tested throughout our operation to ensure there are no adverse health and safety concerns. This is to ensure there is no safety or health risk to NULIFE's team, and the public. Extensive produced gas analysis has been done from laboratory scale tests, through to the existing commercial stage.

All gasses generated in the system have been continually tested to ensure there are no adverse health or environmental effects associated with them. These results are well documented and have been tested by third parties.

Outgoing waste/materials

Effluent Water: The primary outgoing waste material is excess effluent waste water. Our long term strategy is to reuse as much effluent water as possible in our system by blending it with dry biomass to create the HTL inlet slurry. We are also investigating the use of an onsite water treatment system that will allow us to safely dispose of the effluent water into the municipal sewer system. Currently, when the wastewater must be disposed of, it is trucked to a local disposal site for safe disposal using a 3rd party specialist waste contractor.

Water storage and handling includes a 400 barrel tank installed within a containment system on the west side of the exterior yard. The secondary containment system has been sized according to Saskatchewan Directive S-01 “Upstream Petroleum Industry Storage Standards”. The water storage tank is connected to an activated carbon canister.

Release of hazardous materials, chemicals and pesticides

Any chemicals used at the site are used in extremely low dilution levels with water for cleaning purposes. When these cleaning mixtures are used, they are safely contained and are properly and safely disposed of.

Pesticides are not applicable since they are not used.

Noise & Vibrations

Our facility does not have any vibration concerns and the system is silent from outside of the building. We have been operational for +2 years at our Saskatoon site and have not had any complaints related to either of the points. We are zoned in a heavy industrial location which has no limitations for noise, vibrations, and odours.

Injection Site

The injection site underwent an environmental assessment prior to securing the [redacted] permits.

Regular monitoring is undertaken at the facility, including:

- Annual groundwater testing to ensure no groundwater contamination
- When required, lab analysis on all emplaced material to ensure compliance with permitting. All NULIFE deliveries of materials are sampled and stored for 6 months, as per permitting requirements. If the cavern experiences abnormal operational issues, the operator may decide to test samples at their own discretion.
- Monthly lab analysis on brine discharge.
- Monthly reporting on total volume of emplaced material.
- Daily readouts of pressures and stability of the subsurface caverns.

The injection site has scrubbers in place to mitigate air pollutant emissions from tanks onsite. Operators utilize portable gas monitors that test for Lower Explosive Limits (LEL) of combustible gases, Carbon Dioxide (CO), Hydrogen Sulfide (H2S) and oxygen (O2) levels. The plant contains four additional fixed gas monitors that test for LELs, CO, H2S and O2 levels.

Ground water monitoring is in place to assess and mitigate pollutant discharges to water. The facility does not have vibration issues. Noise assessments are done via routine

facility decibel testing. Since the site is a disposal facility, nothing leaves. This minimizes risks related to the release of hazardous materials and wastes. Material cannot be sent elsewhere, as the injection site is the end location. [redacted] permitting requirements create compliance requirements that prevent the improper handling of any waste material.

Compliance with CSA Z341.2 ensures environmental safety by enforcing strict requirements for the containment and isolation of stored materials within salt caverns. The standard mandates detailed site characterization, well integrity verification, and continuous monitoring to prevent the migration of stored substances into surrounding rock formations, groundwater, or the atmosphere. It also specifies stringent closure procedures—including well plugging and surface isolation—to eliminate potential long-term leakage pathways. By addressing both geological and engineered barriers, and requiring operators to adhere to rigorous environmental protection practices, CSA Z341.2 provides a robust framework that protects underground sources of drinking water, surface ecosystems, and surrounding communities from any adverse environmental impacts associated with subsurface storage.

D.2.2 Biodiversity conservation and sustainable management of living natural resources, including terrestrial & marine biodiversity and ecosystems, protecting habitats of rare & endangered species, avoiding conversion of natural forests, grasslands or wetlands, minimizing soil degradation or erosion, minimizing water consumption and stress.

x	Above risks are applicable to this Project
●	Not applicable to this Project

Environmental impacts need to be considered for:

1. HTL Conversion
2. Injection Site

HTL Conversion

We are operating in a preexisting heavy industrial zone within city limits. The old tenant in our building was an industrial hydraulics repair and manufacturing business. NULIFE's Saskatoon facility is not situated near any cultural heritage or religious sites. Our site does not displace valuable habitat or impact any animal species. If you use Google Maps to assess our location (1725 Ontario Avenue, Saskatoon) we have no nearby water or ecologically sensitive lands. The project has no negative impacts on terrestrial and marine biodiversity and ecosystem.

NULIFE is dedicated to reducing soil degradation and erosion to the greatest extent possible. By diverting industrial waste streams from being landfilled or landspread, NULIFE promotes reduction of soil contamination.

Our operations are primarily indoors, which acts as a containment mechanism to prevent negative impacts to soils. The commissioning of our double wall tanks with bern systems highlight our willingness to protect the soil onsite.

The project has not displaced valuable habitat or impacted any animal species.

Our technology utilizes wet waste biomass, allowing us to minimize (or eliminate) water usage. Since HTL is a net generator of water, commercially, we could increase regional water supply.

Injection Site
The salt cavern storage site has been operational since the 1960s with a fixed, small footprint that does not expand, ensuring minimal impact on terrestrial and marine biodiversity, ecosystems, and habitats. The facility does not encroach upon or convert natural forests, grasslands, wetlands, or high conservation value areas, as it was established on previously developed land. Underground storage minimizes surface disruption, preventing soil degradation, erosion, and habitat fragmentation. Additionally, operations do not place significant demand on water resources, avoiding water stress while adhering to strict regulatory requirements under [redacted] to ensure environmental protection and sustainability. As the facility operates under [redacted], it complies with all provincial regulations governing underground storage, including environmental and safety assessments.

D.3 Social Impacts

D.3.1 Labor rights and working conditions, including providing safe & healthy working conditions for employees, fair treatment and equal opportunities in your organization; considerations of prevention of forced labor, child labour or trafficked persons protecting workers employed by third parties.

This is required for all Projects.

x	Above risks are applicable to this Project

NULIFE GreenTech's comprehensive corporate safety programs play a pivotal role in the company's broader social responsibility and community impact framework. Core components such as the *Violence Prevention in the Workplace Policy*, *Harassment Policy*, and *Substance Abuse Policy* contribute to fostering a safe, respectful, and inclusive work environment, supporting the well-being of employees and contractors. These initiatives are critical in minimizing social risks often associated with industrial operations, particularly in proximity to urban or Indigenous communities. Furthermore, protocols like *Hazard Identification and Control*, *Emergency Response Plans*, and *Incident Reporting and Investigation* enhance transparency and trust by ensuring swift mitigation of health and safety issues. The *Environmental Policy* and *Return to Work (RTW) Policy* demonstrate a commitment to both ecological stewardship and worker rehabilitation—key indicators in a social impact assessment. Regular *Safety Training*, *Meetings*, and *Inspections* ensure that both internal and external stakeholders are actively engaged in safety and sustainability, reinforcing NULIFE's role as a responsible employer and community partner.

NULIFE is committed to upholding human rights and avoiding discrimination. Its practices

are designed to ensure respect for all individuals, regardless of race, gender, religion or other factors. Since 2016, NULIFE has provided high-quality employment that also advances equity, diversity, and inclusion. We only do business with other companies and investors who share our same values. NULIFE provides a safe and healthy workplace for all employees and third-party contractors. NULIFE provides work flexibility to all employees to manage work life balance in a style that is suitable to their needs. Other benefits offered to employees include: Paid vacation, competitive wages, flexible employee benefits, employee stock option plan along with personal and professional development opportunities. NULIFE is committed to the creation of high-quality employment opportunities for our local community. NULIFE is regulated and adheres to Canadian employment regulations such as the Workers Compensation Board (WBC).

A key benefit of this project is local job creation including jobs for Indigenous Peoples and University of Saskatchewan graduates. NULIFE co-founders (Brock Eidem and Jerry Kristian) are the majority shareholders and have been local Saskatoon residents for their whole lives. Outside investment from our company has been primarily through Saskatoon angel investors including [redacted]. The success of this project will result in wealth-creation that would stay in the local community and with Indigenous Peoples.

Revenue from the project is used to support and grow operations. We do business with local companies. For example, our welding, electrical and Programmable Logic Controller (PLC) work is all done by local Saskatoon owned companies. As we use carbon credit sales to expand operations, this revenue will be re-deployed back into the local community.

Our third-party injection partner is a licensed business in good standing within the Canadian jurisdiction. Due to Canadian law, they are governed by laws and regulations which stipulate against practices such as forced labour, child labour, and the employment of trafficked persons.

D.3.2 Land acquisition and involuntary resettlement in the context of your deployment site selection

●	Above risks are applicable to this Project
x	Not applicable to this Project

No forced physical and /or economic displacement has occurred for this project. We moved into an preexisting industrial building that was available for lease by the owner. Our industrial area within the city of Saskatoon was developed before 1910. Our building was built decades ago (~1970) and we moved in as a leased tenant in 2023. Local communities were not displaced in the development and execution of this project.

No known involuntary resettlement has occurred due to commercial activities at the injection site since the 1960s.

D.3.3 Environmental and social justice and gender equality (equal opportunities and pay), as it relates specifically to deployment site selection.	
x	Above risks are applicable to this Project
●	Not applicable to this Project

There are no known environmental, social justice and gender inequality risks associated with NULIFE's Saskatoon deployment site.

Diversity, equity, and inclusion is a core value within our company. Since our inception in 2016 we have had a diverse workforce. As we are a small 8 person team we have not had a formal policy. We have always felt our actions speak louder than any words on paper. It is part of our plan for 2025 to develop a formal policy which represents our corporate values and culture regarding diversity, inclusion, and equality. Currently we have 25% of our workforce which meets the criteria of DEI.

We have a signed Letter of Support for our project from [redacted]. NULIFE has continued to build respectful relationships with Indigenous peoples. As our company expands, we will always receive informed consent from Indigenous peoples before proceeding with economic development opportunities that impact Indigenous communities. [redacted] is a minority shareholder of our company, resulting in Indigenous ownership for this project.

There are no known environmental, social justice or gender equality risks that result from third party sites of this project. All sites have been long-established as commercial operations. All third-party partners operate in good standing regarding regulatory requirements for human rights, worker safety and nondiscrimination.

D.3.4 Human rights, and in particular gender-based violence, including sexual harassment, human trafficking, slavery, imprisonment, physical and mental drudgery, and punishment or coercion.	
x	Above risks are applicable
●	Not applicable to this Project

Saskatchewan and Canadian law provide strong legal frameworks to mitigate risks related to human rights violations, including gender-based violence, sexual harassment, human trafficking, slavery, and coercion. Canada upholds international human rights standards through laws such as the *Criminal Code*, *Human Rights Act*, and *Employment Equity Act*, which explicitly prohibit discrimination, forced labour, and workplace harassment. Saskatchewan's *Occupational Health and Safety Regulations* further mandate safe, respectful workplaces, with clear policies to prevent harassment and violence. Strict enforcement, regulatory oversight, and legal recourse ensure that individuals are protected from exploitation, reinforcing Canada's commitment to upholding fundamental human rights.

D.3.5 Confirm whether there are **indigenous land rights, ancestral or customary rights or equivalent claims** pertaining to the area in which the project intends to operate**.** Outline any measures being taken to mitigate any negative impacts on rights-holders, indigenous peoples and local communities.

●	Applicable to this Project (please provide details below)
x	Not applicable to this Project

This project does not negatively impact indigenous peoples in any way nor does the project impose on any land/customary rights. No forced physical and /or economic displacement has occurred and will not occur in the future.

Our current company operates in a manner that considers the three key sections of the Business and reconciliation section of Canada's 92nd Call to Action for the Corporate Sector.

NULIFE respects the National Truth and Reconciliation Day. Our Human Rights policies comply with Canadian and Saskatchewan human rights legislation. NULIFE has a Human Rights Policy to ensure respect for the dignity, rights, and freedoms of all individuals, including a zero-tolerance stance on discrimination, harassment, or abuse, and a commitment to upholding workers' rights, ensuring that all employees are treated equitably. Since its inception in 2016, NULIFE has been a strong advocate for equity, diversity, and inclusion. The company ensures that its employment practices reflect these values and only collaborates with partners who share the same commitment to human rights and ethical business practices. This internal due diligence process includes initial screening, compliance verification, documentation review, risk assessment and approval/monitoring.

As per [redacted], for injection site permitting to be approved, the appropriate measures must be taken to account for any impacted freehold mineral owners and written consent must be received for all areas impacted by the injection site operations.

D.4 Additional risks

In addition to the above areas of environmental and social impacts, I attest that further risks of potential harm arising from the Project have been assessed, and [please select one below option]:

	A) No such additional risks have been identified
X	B) For any such risks identified, appropriate risk mitigations have been put in place and are detailed below.

Please describe any such risks identified, and the corresponding risk mitigations which have been put in place.

No other risks identified.

D.5 Sustainable Development Impact

Please explain in the below table, where relevant, which Sustainable Development Goals (SDGs) other than SDG13 ("Climate Action") your Project has a positive impact and how this is consistent with the SDG objectives of all jurisdictions in which you operate. If applicable, please include information on any data and parameters monitored before or during the crediting period.

Please see here for a full list and details of all 17 SDGs.

#SDG with name	**Description of project impact (***as per selected indicators corresponding to the targets (example)predefined under each SDG by the UN***)**	**Description of alignment with indicators of jurisdiction(s), if relevant and applicable**	**If applicable, unit of measurement (e.g. removal credits, MWh)**
6 - Clean Water & Sanitation	NULIFE's commitment to aligning with SDG 6 (Clean Water and Sanitation) is crucial in the context of emerging environmental health risks associated with the land application of biomass waste such as biosolids. By utilizing challenging waste biomass such as effluent sludges, fats, oils, and greases (FOG) and biosolids, NULIFE effectively mitigates the risk of pollutants impacting local water sources as well as leaching from landfill application.		
8 - Decent work & economic growth	The activities of this project support local economic growth by creating new competitive job opportunities.		
9 - Industry, innovation & Infrastructure	The activities of this project support the development of novel technology and industrial infrastructure to address waste challenges while		

	removing carbon from the atmosphere.		
11- Sustainable Cities and Communities	The activities of this project promote waste diversion, which results in more sustainable operations for our local feedstock suppliers. By requesting renewable electricity from SaskPower, we are sending a signal at the provincial level that there is demand for lower carbon intensity electricity, which creates a cleaner community.		
12 - Responsible consumption & production	Most of the waste managed by NULIFE would otherwise end up in farmland lagoons, landfills or wastewater treatment plants. All three of these solutions pose their own risks due to the level of contaminants within the waste. Farmland lagoons pose risks of leaks of phosphorus and nitrates in local ecosystems, disturbing the biodiversity, while emitting odours unpopular with local communities. Landfills emit methane and toxins in the atmosphere and are exposed to high risks of water systems contamination. Organic waste, including food processing wastes, sewage sludge and bio-solids, contributes significantly to this pollution crisis, prompting regions in North America to prohibit their disposal in landfills such as California or British Columbia. NULIFE's technology allows for thousands of tonnes of waste that can not be valorized to not reach manure lagoons, landfills or wastewater treatment plans. This reduces the release of contaminants into the		

	environment, reducing harm to local ecosystems and communities.		

Please explain whether you assessed the SDG impact of your project prior to this qualitative reasoning. If so, please describe if and which standardized assessment tools and methods you relied on.

Description of assessment tools and methods:

There has not been a formal assessment of SDG impacts – The qualitative content above is based on baseline project assessment by NULIFE.

E – Stakeholder Input Process

E.1 Stakeholder Input Process Summary

Please provide a description and documentation of how comments by local stakeholders have been invited and compiled, a summary of comments received, and report on how due account was taken of comments received. Refer to Section 3.5 "Stakeholder Input Process" of the Isometric Standard for full requirements.

We started the development of our Saskatoon Project in 2022. At the time, Isometric was not yet a company, thus we were unaware of the Isometric Stakeholder Input process.

Up until today, we have engaged with stakeholders via face-to-face conversations or virtual meetings along with on-site tours of our Saskatoon Plant. We continue to tour stakeholders and other interested parties regularly. The overwhelming feedback from all engagements is the support surrounding sustainability, waste reduction and the creation of new employment opportunities.

We regularly tour University of Saskatchewan students at our facility and we often receive outreach from students who are looking for employment opportunities. Almost our entire workforce is University of Saskatchewan graduates.

When deciding on relevant stakeholders to consult, we considered the geographical location of our project, Indigenous Peoples and Local Communities, local policy makers and government officials. During site visits, we note any questions and concerns.

Stakeholders consulted includes:

- The Government of Saskatchewan, since 2017. Comments: Positive feedback related to sustainability and the overall Saskatchewan solid waste management strategy.
- The Government of Canada. In 2024 we toured the Deputy Minister of Agriculture through our Saskatoon Plant.

- Peter Ballantyne Group of Companies (Cree Nation), since 2023. Positive feedback related to sustainability, future collaborations, employment/training opportunities for local Indigenous peoples.
- University of Saskatchewan, since 2017. Positive feedback related to Saskatchewan innovation, waste reduction, decarbonization and hiring University graduates.
- Saskatchewan Ministry of Environment related to site permitting. Engaged since 2018.
- Site neighbours, since 2023. Positive Impacts: We have had friendly relationships with our neighbours. We continue to engage with our site neighbors as a part of our requirements with the Saskatchewan Ministry of Environment.
- Feedstock suppliers, since 2017. Positive Impacts: Sustainable waste management solution for them while reducing their disposal/transport costs.
- City of Saskatoon wastewater treatment plant, since 2022. Positive Impacts: Long term alignment on solutions for their wet biosolids waste.
- City of Saskatoon municipality. Engaged since 2022 for our current facility. The City of Saskatoon provided us with our business license to operate.

Letters of support are attached from The Government of Saskatchewan, Peter Ballantyne Group of Companies and the University of Saskatchewan.

As part of the stakeholder engagement process, we received questions on the impact on air quality from a site neighbour. NULIFE immediately addressed the concerns with the neighbouring building owner to ensure there are no ongoing air quality concerns. NULIFE has engaged in continuous follow-up with the neighbour.

When we receive inquiries about our project, they are documented and addressed instantly.

<u>Injection Site</u>
One of the reasons that we selected to work with our Injection Partner is due to the injection sites longstanding involvement with the local [redacted] community. The injection site has been a staple in its local community since the 1960s, which has resulted in local job creation. NULIFE's work with the Injection partner does not require any significant modification of their operations in a manner that would impact local stakeholders. The Injection Partner works closely with the [redacted]. The permitting body has their own stakeholder input process and grievance mechanism for ongoing concerns and complaints. The complaints are documented and resolved and they occur.

E.2 Grievance Mechanism

Please outline the mechanism for stakeholders to voice, process and resolve grievances.

We are a team of < 10 employees in a small local community. Our office door is always open and we welcome visitors. Our phone number is accessible on a Google search and

we have a ‘contact us’ box on our website along with our phone number. I (Kieran Lewko) monitor our website’s “contact us box” and have the ability to respond immediately to any inquiry. This timeline is much faster than the stated 14 day timeline.

Due to the size of our team, we can typically process grievances efficiently and have them resolved in under 60 days.

NULFE has an open communications channel with Saskatchewan regulators and any grievances presented to them are addressed on a case by case basis.

Injection site: The [redacted]s has their own stakeholder input process and grievance mechanism for ongoing concerns and complaints. The complaints are documented and resolved and they occur.

Section F – Project Changes

F.1 Summary of changes

Please provide details of the proposed or actual changes to the implementation, operation or monitoring of the project activity description submitted to Isometric.
Project Proponents must determine whether the proposed or actual changes are temporary deviations or permanent changes

If there are temporary changes or deviations which are only relevant to a specific Reporting Period, please note this instead in Section C in the corresponding GHG Statement Report.

(Kindly note that this section may not be applicable during the initial validation. If so, kindly leave the section blank and mention "Section is intentionally left blank.")

F.2 Impact of changes

Please provide details on the impact of the proposed or actual changes (temporary/permanent) on the following sections of the project:

a) *Additionality*
b) *Applicability of Protocol/ Module*
c) *Quantification of the carbon removals*
d) *Any other (stakeholder consultation, safeguard assessment etc.)*

(Kindly note that this section may not be applicable during the initial validation. If so, kindly leave the section blank and mention "Section is intentionally left blank.")

N/A

F.3 History of changes

Please provide the summary of the changes in the table
(Kindly note that this section may not be applicable during the initial validation. If so, kindly leave the section blank and mention "Section is intentionally left blank.")

Date of change	*Nature of change*	*Impact of change*	*Corresponding version of the PDD*

Document History		
Version	**Date**	**Description**
v1.7	August 2024	B.11 Reversals updated in line with Isometric Standard questionnaire update Inclusion of D3.4 and D3.5
v1.6	June 2024	Clarification to explanation that “B.14 Storage site overview” is only applicable to closed systems; revision to require “Regulatory Authority / Jurisdiction” to be provided for this question
v1.5	May 2024	Revision to include section on project changes, guidance to the template and other editorial changes
v1.4	November 2023	Public version (initial adoption)

Biomass Feedstock Accounting PDD Appendix

Version 1.1

This Appendix must be completed for each feedstock type and feedstock provider used in this Project.

A1. Feedstock Summary

Please describe your Feedstocks used.

The project relies on local waste biomass from the Saskatoon & area.

Feedstock 1: [redacted] provides us with effluent sludge from their [redacted] Canola Crushing Plant which is 30km away from NULIFE’s site. The sludge is an emulsion with high variability and moisture levels. The sludge is derived from their Dissolved Air Flotation (DAF) system along with other high moisture waste generated from activities such as washing out railway cars and plant spills. [redacted] currently has a [redacted] of sludge that they cannot get rid of along with ongoing waste generation. They have explored all viable options including biofuels. It is not viable for Anaerobic Digestion (AD), nor are there any viable AD units in Saskatchewan. The reason they are stock piling it is because they had no cost-effective disposal route. The stockpile can be validated by supplier documentation and a picture of the stockpile.

Feedstock 2: Oat Hulls consist primarily of the outer covering of oats, obtained in the milling of table cereals or in the grouting of oats from clean oats. Oat Hulls, due to their fluffy nature, do not flow well and can create handling problems in improperly designed mechanical systems. Oat hulls will blow away in the wind if not contained properly and are a fire risk when stockpiled.

Oat hulls are used as cattle feed/animal bedding and then disposed of (e.g., spread in the field). This carbon re-enters the atmosphere after it is consumed or after decomposition in the field. NULIFE will demonstrate that there is an oversupply of oat hulls in the region for a replacement emissions exemption.

Oat Hulls Supplier 1: [redacted] provided us with oat hulls from their Saskatoon location that is 24 km from the NULIFE plant. The oat hulls are a by-product from their oat milling process.

Oat Hulls Supplier 2: [redacted] provides us with oat hulls from their [redacted] location that is 40 km from the NULIFE plant. The oat hulls are a by-product from their oat milling process.

Feedstock 3: Waste Bran. [redacted] us with their waste oat bran from their oat milk production plant in Saskatoon, which is 9km away from NULIFE's site. The baseline scenario is landfilling into a local landfill.

Current and future feedstocks will always undergo the Isometric Biomass Feedstock accounting module screening analysis for eligibility.

A2. Feedstock Hazardous status

Please describe how you are demonstrating that the feedstock you are using is not hazardous. This may either be done by providing evidence from tests of the feedstock or by providing evidence that the relevant injection permits only allow non-hazardous materials.

We are processing industrial waste streams that are classified as non-hazardous. By engaging with feedstock suppliers, we can determine if the feedstock is classified as hazardous or not.

We are not injecting biomass. The HTL process converts the biomass into new products, such as bio-oil. Our bio-oil is confirmed non-hazardous, as per the SDS and TCLP testing disclosed in Section D2 of the PDD.

A3. Feedstock Provider Organizations

Please provide a complete list of organizations involved in providing the feedstock, clarifying the organization's role and providing contact information for each.
(*Please duplicate the below rows for each additional organization you wish to add)*

Organization 1

Company registration number (*Unique business identification number in your country of registration*): Federal Business Number (BN): [redacted]
Organization Name: [redacted]
Organization Address: [redacted]
Contact Person: [redacted]
Contact Email Address: [redacted]
[Optional] Contact Phone Number: [redacted]
Organization role in project: Feedstock supplier

A4. Potential Market Leakage Impacts Eligibility

Please select which of the Potential Market Leakage Impact Eligibility Criteria you're using to demonstrate eligibility for this feedstock.

To establish eligibility under market leakage criteria, the Project Proponent must demonstrate either:

- *The feedstock meets at least one of EC1 through EC4; or*
- *The feedstock meets at least one of EC5 through EC7 and the feedstock meets at least one of EC8 through EC12.*

[Yes]	**EC1: Project Proponent does not pay for the feedstock used**
[Yes/No]	EC2: Project proponent is paid a "tipping fee" to remove the feedstock
[Yes/No]	EC3: Project Proponent pays for recovery & replacement activities only
[Yes/No]	EC4: Project Proponent pays to a 3rd party, not entity producing feedstock

[Yes/No]	EC5: Feedstock is a forest residue
[Yes/No]	EC6: Feedstock is an agricultural crop residue or an animal waste
[Yes/No]	EC7: Feedstock is a non-marketable industrial waste
[Yes/No]	EC8: Biomass originates from a regulated forest management project
[Yes/No]	EC9: Certified forest management activity in increasing carbon stock areas
[Yes/No]	EC10: Certified forest management activity in exceptional circumstances
[Yes/No]	EC11: Sustainably sourced agricultural crop residue
[Yes/No]	EC12: Surplus residue with no demonstrated growth of supply
Demonstrate how your feedstock meets the criteria by providing the required documentation or analysis.	
EC1 conditions will be satisfied by documentation disclosing the cost neutral feedstock between NULIFE and [redacted]. Certification of waste acceptance document between Nulife/[redacted] uploaded as evidence. Attachment: 2025-05-26 [redacted] Packing Slip and certificate of waste acceptance EC1 Isometric compliance.pdf	

A5. Counterfactual Storage Eligibility

Please select which of the Counterfactual Storage Eligibility Criteria you're using to demonstrate eligibility for this feedstock.

[Yes]	**EC13: Project Proponent must assess the counterfactual fate of the feedstock**
[Yes/No]	EC14: Government managed wildfire mitigation or restoration activity

Demonstrate how your feedstock meets the criteria by providing the required documentation or analysis.

EC13: A qualitative assessment that the expected fate of the biomass would have a durability lower than the threshold given the most economically viable option in a given Sourcing Area.

Since the biomass has no viable cost effective disposal route,and the supplier is stockpiling, the counterfactual storage eligibility is selected to be landfilling, which generates CO2 and CH4 emissions during decomposition. NULIFE has engaged with Isometric on the counterfactual storage eligibility scenario and landfilling was determined to be the appropriate scenario.

Saskatchewan is not required to install methane gas capture on landfills: Reducing methane emissions from Canada's municipal solid waste landfills: discussion paper - Canada.ca

The most likely destination for the feedstock in the counterfactual/baseline scenario would be landfilling. The Loraas North Location is a nearby landfill. This landfill is located on Lutheran Road, ½ Mile East of Highway 12 North in the RM of Corman Park. The landfill does not have gas capture technology. This can be validated by the attached thesis: "2021 Thesis Loraas North Landfill". Section 1.2.1 Site Location and Background explains that the site could be a candidate for methane gas capture but it currently does not have it in place.

The landfill decay rate utilized is 19% per year via Table 1 of the following EPA source: Quantifying Methane Emissions from Landfilled Food Waste.

See attachment: 2025-07-03 NULIFE Feedstock Counterfactual Calculations

A6. Dedicated Energy Feedstock Eligibility

This is only applicable to non-forestry feedstocks.

EC15: Feedstock is not used for energy production

Demonstrate how your feedstock meets the criteria by providing the required documentation or analysis.

EC15 is not relevant to this feedstock and not applicable to this project. The sludge is derived from their Dissolved Air Flotation (DAF) system along with other high moisture waste generated from activities such as washing out railway cars and plant spills.

A7. Counterfactual Fate of Feedstock

Please describe and attach any relevant evidence to demonstrate what the most likely counterfactual scenario would be for your feedstock, using guidance outlined in Section 3.2.1 of the Biomass Feedstock Module.

[redacted] currently has a [redacted] of sludge that they cannot get rid of. They have explored all viable options including biofuels. The reason they are stock piling it is because they had no cost-effective disposal route. The counterfactual fate of the sludge can be validated by the supplier letter in the appendix plus a picture of the stockpile.

Attachment: 2025-05-22 Signed - NULIFE [redacted] FOG Biomass Feedstock Isometric Standard Ownership Compliance Letter RW.pdf

A8. Replacement Emissions

Please select which method of replacement emissions accounting you've selected given the nature of your feedstock	
[Yes]	**Accounting for the feedstock replacement emissions in the LCA**
[Yes/No]	Not accounting for replacement emissions due to exemption C1 *(Feedstock has no counterfactual use)*
[Yes/No]	Not accounting for replacement emissions due to exemption C2 *(Feedstock counterfactual use is most likely replaced with a feedstock with no counterfactual use)*
[Yes/No]	Not accounting for replacement emissions due to exemption C3 *(Feedstock has no counterfactual use due to surplus)*
If you selected an exemption above, please demonstrate how your feedstock meets it by providing the required documentation or analysis.	
C1 exemption due to stock pile evidence via supplier documentation and picture of the stockpile. Attachment: 2025-05-22 Signed - NULIFE [redacted] FOG Biomass Feedstock Isometric Standard Ownership Compliance Letter RW.pdf Attachment: 2025-05-22 [redacted] Evidence Photo.jpg	

A3. Feedstock Provider Organizations
Please provide a complete list of organizations involved in providing the feedstock, clarifying the organization's role and providing contact information for each. (*Please duplicate the below rows for each additional organization you wish to add)*
Organization 2a (under development)
Company registration number (*Unique business identification number in your country of registration*): [redacted]
Organization Name: [redacted]
Organization Address: [redacted]
Contact Person: [redacted]
Contact Email Address: [redacted]
[Optional] Contact Phone Number: [redacted]
Organization role in project: Feedstock supplier

A3. Feedstock Provider Organizations
Please provide a complete list of organizations involved in providing the feedstock, clarifying the organization's role and providing contact information for each. (*Please duplicate the below rows for each additional organization you wish to add)*
Organization 2b (under development)
Company registration number (*Unique business identification number in your country of registration*): [redacted]
Organization Name: [redacted]
Organization Address: [redacted]
Contact Person: [redacted]
Contact Email Address: [redacted]
[Optional] Contact Phone Number: [redacted]
Organization role in project: Feedstock supplier

A4. Potential Market Leakage Impacts Eligibility

Please select which of the Potential Market Leakage Impact Eligibility Criteria you're using to demonstrate eligibility for this feedstock.

To establish eligibility under market leakage criteria, the Project Proponent must demonstrate either:

- *The feedstock meets at least one of EC1 through EC4; or*
- *The feedstock meets at least one of EC5 through EC7 and the feedstock meets at least one of EC8 through EC12.*

[Yes/No]	EC1: Project Proponent does not pay for the feedstock used
[Yes/No]	EC2: Project proponent is paid a "tipping fee" to remove the feedstock
[Yes/No]	EC3: Project Proponent pays for recovery & replacement activities only
[Yes/No]	EC4: Project Proponent pays to a 3rd party, not entity producing feedstock
[Yes/No]	EC5: Feedstock is a forest residue
[Yes]	**EC6: Feedstock is an agricultural crop residue or an animal waste**
[Yes/No]	EC7: Feedstock is a non-marketable industrial waste
[Yes/No]	EC8: Biomass originates from a regulated forest management project
[Yes/No]	EC9: Certified forest management activity in increasing carbon stock areas

[Yes/No]	EC10: Certified forest management activity in exceptional circumstances
[Yes]	**EC11: Sustainably sourced agricultural crop residue**
[Yes/No]	EC12: Surplus residue with no demonstrated growth of supply

Demonstrate how your feedstock meets the criteria by providing the required documentation or analysis.

EC6: Feedstock is an agricultural crop residue or an animal waste.

NULIFE will demonstrate that the feedstock is an agricultural residue or waste whose production is ancillary to the production of the primary marketable product. Evidence will be provided via supplier documentation.

EC11: Compliance after discussion with Isometric.

The feedstock is an industrial biomass waste rather than an agricultural waste obtained directly from a grower; as the oats are the primary biomass product and the oat hulls are a biomass waste product from the processing to obtain the primary product. NULIFE won't be able to compel the plant owner to obtain the evidence for EC11 from the farmers as there are a large number oat farmers in the supply chain with different practices, and the processing plant has minimal incentive to obtain this information.

EC6 and EC11 are in compliance.

Attachment: Will be submitting with an upcoming GHG Statement Report when the feedstock is utilized.

A5. Counterfactual Storage Eligibility

Please select which of the Counterfactual Storage Eligibility Criteria you're using to demonstrate eligibility for this feedstock.

[Yes]	**EC13: Project Proponent must assess the counterfactual fate of the feedstock**
[Yes/No]	EC14: Government managed wildfire mitigation or restoration activity

Demonstrate how your feedstock meets the criteria by providing the required documentation or analysis.

EC13: A qualitative assessment that the expected fate of the biomass would have a durability lower than the threshold given the most economically viable option in a given Sourcing Area.

If oat hulls are either used as animal bedding/feed and then landfilled or disposed of (e.g., spread in the field). This carbon re-enters the atmosphere after disposal or from decomposition in the field.

Documentation from oat milling suppliers show that these materials would otherwise have been left exposed to decay on site or be transported off site and remain unburied, decaying over time. That these materials would remain unburied and exposed to normal weathering conditions implies that no additional carbon sequestration into soil or other natural sink would occur. Furthermore, literature review supports the claim that the materials used in NULIFE's process – oat hulls – fully decay in outdoor conditions within the 15-year time frame cited in the eligibility criteria EC13 and equations 1 and 2 in section 2.2 of the Biomass Feedstock Accounting module. Therefore, the baseline to which carbon removals are being compared is a net zero sequestration counterfactual. In other words, the baseline we subtract from our calculation of net CO2e removal is zero.

See attachment: 2025-07-03 NULIFE Feedstock Counterfactual Calculations

A6. Dedicated Energy Feedstock Eligibility

This is only applicable to non-forestry feedstocks.

EC15: Feedstock is not used for energy production

Demonstrate how your feedstock meets the criteria by providing the required documentation or analysis.

EC15: Biomass was not grown for the purpose of energy production. It is a plant by-product.

A7. Counterfactual Fate of Feedstock

Please describe and attach any relevant evidence to demonstrate what the most likely counterfactual scenario would be for your feedstock, using guidance outlined in Section 3.2.1 of the Biomass Feedstock Module.

Evidence will be provided from the feedstock supplier outline the past use of their biomass.

Attachment: Will be submitting with an upcoming GHG Statement Report when the feedstock is utilized.

A8. Replacement Emissions

Please select which method of replacement emissions accounting you've selected given the nature of your feedstock

[Yes/No]	Accounting for the feedstock replacement emissions in the LCA
[Yes/No]	Not accounting for replacement emissions due to exemption C1 *(Feedstock has no counterfactual use)*
[Yes/No]	Not accounting for replacement emissions due to exemption C2 *(Feedstock counterfactual use is most likely replaced with a feedstock with no counterfactual use)*
[Yes]	**Not accounting for replacement emissions due to exemption C3** ***(Feedstock has no counterfactual use due to surplus)***

If you selected an exemption above, please demonstrate how your feedstock meets it by providing the required documentation or analysis.

Replacement emissions are 0 due to C3 Exemption due to surplus of material.

C3: *For project feedstock usage above the percentage of what is theoretically possible to use for a given purpose (the 'Sustainable Use Rate') the feedstock would be considered true waste and therefore not require calculation of replacement emissions.*

See Attachment: 2025-04-16 Biomass Feedstock Accounting Module - Oat Hulls C3 Replacement Emissions Exemption.pdf

A3. Feedstock Provider Organizations

Please provide a complete list of organizations involved in providing the feedstock, clarifying the organization's role and providing contact information for each.
(*Please duplicate the below rows for each additional organization you wish to add)*

Organization 3

Company registration number (*Unique business identification number in your country of registration*): [redacted]

Organization Name: [redacted]
Organization Address: [redacted]
Contact Person: [redacted]
Contact Email Address: [redacted]
[Optional] Contact Phone Number: [redacted]
Organization role in project: Feedstock supplier

A4. Potential Market Leakage Impacts Eligibility

Please select which of the Potential Market Leakage Impact Eligibility Criteria you're using to demonstrate eligibility for this feedstock.

To establish eligibility under market leakage criteria, the Project Proponent must demonstrate either:

- *The feedstock meets at least one of EC1 through EC4; or*
- *The feedstock meets at least one of EC5 through EC7 and the feedstock meets at least one of EC8 through EC12.*

[Yes]	**EC1: Project Proponent does not pay for the feedstock used**
[Yes/No]	EC2: Project proponent is paid a "tipping fee" to remove the feedstock
[Yes/No]	EC3: Project Proponent pays for recovery & replacement activities only
[Yes/No]	EC4: Project Proponent pays to a 3rd party, not entity producing feedstock
[Yes/No]	EC5: Feedstock is a forest residue

[Yes/No]	EC6: Feedstock is an agricultural crop residue or an animal waste
[Yes/No]	EC7: Feedstock is a non-marketable industrial waste
[Yes/No]	EC8: Biomass originates from a regulated forest management project
[Yes/No]	EC9: Certified forest management activity in increasing carbon stock areas
[Yes/No]	EC10: Certified forest management activity in exceptional circumstances
[Yes/No]	EC11: Sustainably sourced agricultural crop residue
[Yes/No]	EC12: Surplus residue with no demonstrated growth of supply

Demonstrate how your feedstock meets the criteria by providing the required documentation or analysis.

EC1 conditions will be satisfied by documentation disclosing the cost neutral feedstock between NULIFE and [redacted].

Feedstock packing slip with cost neutral evidence uploaded as evidence.

Attachment: 2025-05-23 PRTC PACKING SLIP - Nu Life May 23rd (002)[redacted] Cost Neutral Evidence EC1

A5. Counterfactual Storage Eligibility

Please select which of the Counterfactual Storage Eligibility Criteria you're using to demonstrate eligibility for this feedstock.

[Yes]	**EC13: Project Proponent must assess the counterfactual fate of the feedstock**
[Yes/No]	EC14: Government managed wildfire mitigation or restoration activity

Demonstrate how your feedstock meets the criteria by providing the required documentation or analysis.

EC13: A qualitative assessment that the expected fate of the biomass would have a durability lower than the threshold given the most economically viable option in a given Sourcing Area.

The counterfactual fate of the material is landfilling, which generates CO2 and CH4 emissions during decomposition.
Saskatchewan is not required to install methane gas capture on landfills: Reducing methane emissions from Canada's municipal solid waste landfills: discussion paper - Canada.ca

The most likely destination for the feedstock in the counterfactual/baseline scenario would be landfilling. The Loraas North Location is a nearby landfill. This landfill is located on Lutheran Road, ½ Mile East of Highway 12 North in the RM of Corman Park. The landfill does not have gas capture technology. This can be validated by the attached thesis: "2021 Thesis Loraas North Landfill". Section 1.2.1 Site Location and Background explains that the site could be a candidate for methane gas capture but it currently does not have it in place.

The landfill decay rate utilized is 19% per year via Table 1 of the following EPA source: Quantifying Methane Emissions from Landfilled Food Waste.

Documentation will be uploaded to Certify.

See attachment: 2025-07-3 NULIFE Feedstock Counterfactual Calculations

A6. Dedicated Energy Feedstock Eligibility

This is only applicable to non-forestry feedstocks.

EC15: Feedstock is not used for energy production

Demonstrate how your feedstock meets the criteria by providing the required documentation or analysis.

EC15: Biomass was not grown for the purpose of energy production. It is a plant waste stream. EC15 is not relevant.

A7. Counterfactual Fate of Feedstock

Please describe and attach any relevant evidence to demonstrate what the most likely counterfactual scenario would be for your feedstock, using guidance outlined in Section 3.2.1 of the Biomass Feedstock Module.

Evidence can be provided by documentation from the supplier stating the previous counterfactual fate of the material.

2025-05-26 Feedstock [redacted] Isometric Standard Ownership Compliance Letter.pdf

A8. Replacement Emissions

Please select which method of replacement emissions accounting you've selected given the nature of your feedstock

[Yes/No]	Accounting for the feedstock replacement emissions in the LCA

[Yes]	**Not accounting for replacement emissions due to exemption C1** ***(Feedstock has no counterfactual use)***
[Yes/No]	Not accounting for replacement emissions due to exemption C2 *(Feedstock counterfactual use is most likely replaced with a feedstock with no counterfactual use)*
[Yes/No]	Not accounting for replacement emissions due to exemption C3 *(Feedstock has no counterfactual use due to surplus)*
If you selected an exemption above, please demonstrate how your feedstock meets it by providing the required documentation or analysis.	
Bran waste was previously landfilled, thus replacement emissions are 0 due to C1 exemption. Attachment 2025-05-26 Feedstock [redacted] Isometric Standard Ownership Compliance Letter.pdf	

Document History

Version	Date	Description
v1.1	December 2024	Updates to ECs as part of the v1.2 Feedstock accounting module
v1.0	November 2023	Initial version